

Cite this: *Nanoscale*, 2022, **14**, 13950

Biosynthesized metal oxide nanoparticles for sustainable agriculture: next-generation nanotechnology for crop production, protection and management

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The current agricultural sector is not only in its most vulnerable state but is also becoming a threat to our environment due to expanding population and growing food demands along with worsening climatic conditions. In addition, numerous agrochemicals presently being used as fertilizers and pesticides have low efficiency and high toxicity. However, the rapid growth of nanotechnology has shown great promise to tackle these issues replacing conventional agriculture industries. Since the last decade, nanomaterials especially metal oxide nanoparticles (MONPs) have been attractive for improving agricultural outcomes due to their large surface area, higher chemical/thermal stability and tunable unique physicochemical characteristics. Further, to achieve sustainability, researchers have been extensively working on ecological and cost-effective biological approaches to synthesize MONPs. Hereby, we have elaborated on recent successful biosynthesis methods using various plants/microbes. Furthermore, we have elucidated different mechanisms for the interaction of MONPs with plants, including their uptake/translocation/internalization, photosynthesis, antioxidant activity, and gene alteration, which could revolutionize crop productivity/yield through increased nutrient amount, photosynthesis rate, antioxidative enzyme level, and gene upregulations. Besides, we have briefly discussed about functionalization of MONPs and their application in agricultural-waste-management. We have further illuminated recent developments of various MONPs ($\text{Fe}_2\text{O}_3/\text{ZnO}/\text{CuO}/\text{Al}_2\text{O}_3/\text{TiO}_2/\text{MnO}_2$) as nanofertilizers, nanopesticides and antimicrobial agents and their implications for enhanced plant growth and pest/disease management. Moreover, the potential use of MONPs as nanobiosensors for detecting nutrients/pathogens/toxins and safeguarding plant/soil health is also illuminated. Overall, this review attempts to provide a clear insight into the latest advances in biosynthesized MONPs for sustainable crop production, protection and management and their scope in the upcoming future of eco-friendly agricultural nanotechnology.

Received 18th July 2022,
Accepted 19th August 2022
DOI: 10.1039/d2nr03944c
rsc.li/nanoscale

1. Introduction

A productive and sustainable agricultural system is essential to the existence of humanity. The global population is growing day by day rapidly, and it is forecasted to reach up to about 9.6 billion by 2050; thus, there is an increasing demand for agricultural production (more than 70% of grain production by 2050).¹ Although current agricultural practices can produce enough food for nearly eight billion people,² they cannot continue to do so sustainably without technological intervention.

Deprived environmental and cost efficiency of pesticide use and crop losses due to disease and drought result in significant damages to embodied energy and water and cascading ecological and economic impacts. This signifies a need to launch a new era for agriculture where engineered nanoparticles (NPs) can guide this shift.

Nanotechnology is a pioneering discipline of science that deals with the transformation of materials on a scale of 1 to 100 nanometers (nm).³ Nanomaterials have tremendous potential to address the rising demand for agriculture because of their large surface area-to-volume ratio and unique physicochemical features like size, shape, morphology, dispersibility/solubility, absorptivity, and reactivity. With the help of cutting-edge nanomaterials-based nanotechnology, modern agriculture is evolving into precision agriculture, providing the most output from the available resources. Moreover, nanomaterials with numerous shapes such as nanospheres, nanocubes,

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nanotubes, and nanowires, have distinctive chemical, electrical, magnetic, and optical characteristics,⁴ allowing higher sensitivity, lower detection limits, and faster reaction times.^{5,6} Fertilizers are needed to raise crop output, but they can reduce soil fertility by disrupting the mineral balance.⁷ Moreover, commonly applied pesticides and fertilizers can readily flow away. Besides, the production expense of these agrochemicals is quite high and must be regulated. Therefore, the use of NPs in agriculture majorly seeks to (i) reduce nutrient losses; (ii) boost plant growth/yields; (iii) diminish the number/dosage of agrochemicals used for plant protection; and (iv) minimize production costs,⁸ while optimizing overall agricultural output.

Recently metal oxide nanoparticles (MONPs) have received a great deal of attention from researchers and are regarded as one of the most prospective and outstanding agricultural agents due to their exceptional properties such as high chemical/thermal stability, effective surface properties, flexible pore size, and increased pore volumes.⁹ Numerous MONPs have been extensively investigated for their long-term significance in plant growth and crop productivity. The physical and chemical features of MONPs have been demonstrated to have a crucial impact on their absorption, accumulation, and translocation in various crop species, which has proved beneficial to overall agricultural development/output. Another intriguing and valuable issue is the use of MONPs to prevent and control plant pests and diseases due to their improved efficacy, durability, and, in particular, their large specific surface area, which can promote interactions with live plant cells.

Though MONPs can be prepared *via* chemical and physical processes, they are costly, less eco-friendly, and consume a lot of energy. Hence, biological synthesis approaches incorporating various plant extracts, enzymes, or microorganisms, including bacteria, algae, and fungus, have been proposed as

viable, environmentally benign alternatives to chemical/physical procedures.^{10–12} Moreover, biosynthesis of MONPs has lately gained prominence due to their nontoxic, biocompatible, and environmental-friendly nature.^{13,14} Additionally, biosynthesized MONPs offer various advantages, such as lower energy consumption and less reliance on harmful chemicals.^{15,16}

Previously, we have reported the benefits of nanomaterials over traditional agrochemicals.⁶ In this review, we have mainly focussed on biosynthesized MONPs for sustainable agricultural applications. Hereby, we have discussed the biosynthesis of various MONPs (such as Fe₂O₃, ZnO, CuO, Al₂O₃, TiO₂, and MnO₂) using microbes and plant extracts. We have elucidated the impacts/effects of MONPs on various plants with possible mechanisms associated with their interaction, including uptake, translocation, internalization, photosynthesis, antioxidant activity, and gene regulations. Besides, we have illuminated the positive effects of reactive oxygen species (ROS) to further understand the ambiguous impacts of MONPs on plants *via* ROS generation. Moreover, we have discussed about surface functionalization of MONPs along with their application in agricultural waste management and further deliberated the most recent developments of biosynthesized MONPs which have been significantly employed in sustainable agriculture as nanofertilizers (NFZs), nanopesticides (NPDs), nano antimicrobial agents (AMBs), and nanobiosensors (NBSSs) for (i) plants' growth and crop productivity enhancement, (ii) pest/disease management, and (iii) detecting nutrients/pathogens/toxins to safeguard plant and soil health. Furthermore, we have demonstrated the potential of biosynthesized MONPs to accomplish sustainable crop production, protection, and management (as shown in Scheme 1). With this study, we aim to encourage more researchers to develop biosynthesized MONPs based nanoproducts for sustainable agriculture. Finally, we have irradiated the current challenges associated with bio-



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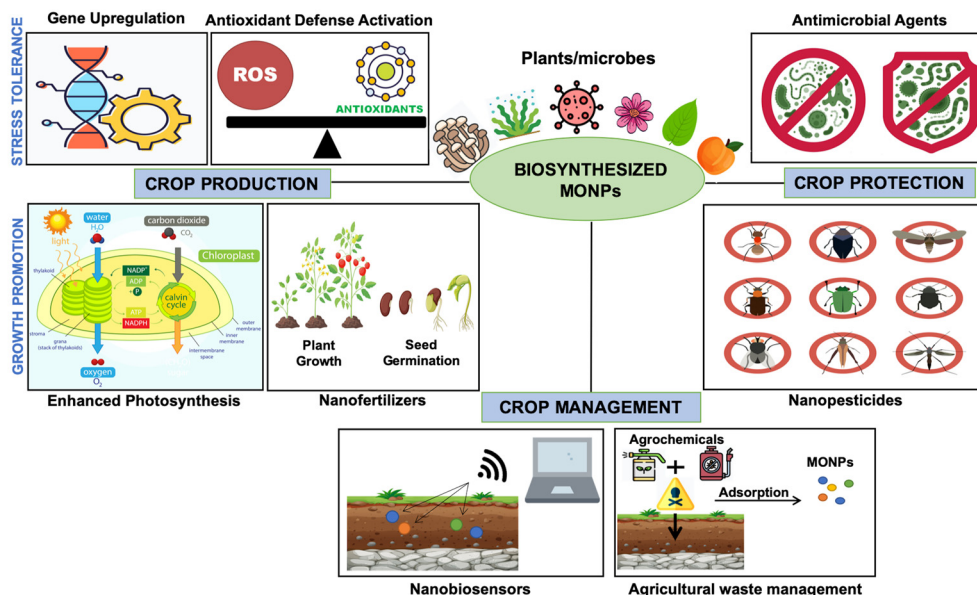
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Scheme 1 Application of biosynthesized MONPs as nanofertilizers, nanopesticides, antimicrobial agents, nanobiosensors and for agricultural waste management to achieve sustainable crop production, protection, and management. Biosynthesized MONPs can also induce stress tolerance and further promote plant growth via various mechanisms like gene upregulation, antioxidant defence activation and enhanced photosynthesis.

synthesized MONPs and their future outlook in sustainable agriculture.

2. Nanomaterials in agriculture

Even though nanotechnology is still a relatively new field of agricultural research, the usage of NPs commenced in agricultural operations at the beginning of the twenty-first century to



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enhance soil fertility and crop production, primarily under deprived conditions.^{17,18} Several NPs with promising contributions to alter the agriculture sector have been developed, with their pros and cons.^{18,19} They are extremely capable of improving the quality and safety of food, crop production, and environmental factors.^{20,21} Additionally, NPs are extensively explored to address agriculture-related difficulties (e.g., pollution monitoring, underlying soil problems, crop diseases, pests and pathogens, pesticides, fertilizer, nutrient delivery, and genetic materials).^{22,23} Different types of NPs and their synthesis methods, properties and agricultural applications are listed in Table 1. Moreover, due to the discrete properties of MONPs like very large surface to volume ratio, high thermal/chemical stability, effective surface properties/characteristics, flexible pore size, and high pore volumes, they are widely being researched to alleviate the current challenges of conventional formulations like non-target action, poor solubility, and ecosystem toxicity. The most commonly used MONPs and their biosynthesis approaches are discussed in the following sections.

2.1. Metal oxide nanoparticles (MONPs) in agriculture

Significant research is going on, on the application of MONPs in agriculture due to their unique characteristics, as discussed above. We can find a lot of research work on metal oxides like CeO₂, Fe₂O₃, In₂O₃, Al₂O₃, CuO, La₂O₃, ZnO, ZrO₂, MgO, TiO₂, SnO₂, Cu₂O for their contribution towards sustainable agriculture. Amongst these, iron (Fe), zinc (Zn), titanium (Ti), copper (Cu), aluminium (Al), and manganese (Mn) are the most common ones due to their relatively lower toxicity.

Table 1 Different types of NPs and their synthesis, properties and applications in agriculture

Type of NPs	Examples	Synthesis & properties	Applications	Ref.
Carbon-based	Fullerenes, carbon nanotubes, graphene and its derivatives, graphene oxide, nanodiamonds, and carbon-quantum dots	Synthesized by thermal CVD method	Nano-biosensors	198–200
		High flexibility, rigidity, and tensile strength Size: 0.4–40 nm	Pesticides Irrigation Seed germination Early plant-growth Nano-fertilizes	
Metal-based	Quantum dots	Synthesized by thermolysis CRM & biomedical methods	Nano-pesticides	20, 199 and 201
	Nanogold	Reduced size & high selectivity against bacteria Size: 1–100 nm		
Ceramic-based	Nanosilver and metal oxides	Synthesized <i>via</i> heat and successive cooling	Nano-fertilizers	199, 200 and 202
	SiO ₂ Al ₂ O ₃	Found in amorphous, polycrystalline, dense, porous, or hollow forms Size: 5–270 nm	Nano-pesticide Nano-carriers	
Lipid-based	TiO ₂ ZrO ₂ Liposomes	Synthesized by melted lipid molecules and active ingredients, dispersed in an aqueous medium containing an emulsifier	<i>In vitro</i> evaluation for kinetics, herbicidal activity, and cytotoxicity.	199, 203 and 204
	Solid lipid NPs	Good physicochemical stability and high encapsulation efficiencies Size: 1–100 nm		
Polymer-based	Chitosan, polylactic acid	Synthesized by cascade reactions involving either convergent or divergent approaches	Aimed at pest control and increasing the yield of crops	199, 203 and 205
	Polyglutamic acid Poly(lactic-glycolic acid)	High structural & chemical homogeneity Size: 1–1000 nm		

2.1.1. Iron oxide (Fe₂O₃/Fe₃O₄). Iron oxide NPs (IONPs, particularly magnetite (Fe₃O₄) and maghemite (Fe₂O₃)) are widely used in agriculture; however, they are primarily used in fertilizers and plant development. Fe₂O₃ NPs are recognized as a core of the agriculture business due to their exceptional quantum confinement, high surface energy, improved surface-area-to-volume ratio, and catalytic capabilities.²⁴ Fe is a vital component for plant growth and a cofactor of several enzymes that accelerate plant progression. It is also utilized as an antibiotic in some situations owing to membrane rupture caused by the production of ROS, that results in cell death.²⁵ Moreover, Fe₃O₄ and Fe₂O₃ NPs that are applied to common beans showed an immense effect on the plant's growth and significantly increased crop's yield.²⁶ Nevertheless, Fe₂O₃ NPs are also used as pesticides on *E. coli*, *S. aureus*, and *Anopheles subpictus*, which showed effective antibacterial activities.²⁷

2.1.2. Zinc oxide (ZnO). Zn is the only metal that appears in all six enzyme types. It is an essential micronutrient and has a crucial function in many integral metabolic processes,²⁸ which is why it is so important and extensively utilized. However, it is mostly employed in disease treatment as an antibiotic or even an antibacterial agent. ZnO NPs have received the most attention due to their unique electrical, optical, and therapeutic features. Because ZnO NPs are extremely biocompatible and have a rapid electron transport kinetics rate, they are suited for using as a biological membrane or for other biological applications.²⁹ In a recent study, ZnO NPs are used to grow and develop common beans.²⁶ In another investigation,

they are used as insecticides and exhibited a progressive increase in mortality of *S. oryzae* and *S. cerealella* adults subjected to treated wheat grains by increasing exposure intervals and concentration.³⁰

2.1.3. Titanium oxide (TiO₂). TiO₂ NPs are widely used for AMBs, NFZs, and the decomposition of organic materials in the wastewater.³¹ Since plants can only utilize 2–4% of the energy from sunlight, this is a significant problem. Numerous studies have proven the use of TiO₂ NPs to accelerate the kinetics of photosynthesis because of their intrinsic photocatalytic characteristics. The foliar application of TiO₂ NPs resulted in improved plant growth, increased fruit output, enhanced Chlorophyll (Chl) content, and reduced water stress.³² The effect of TiO₂ was measured for the growth and productivity of common bean which showed good results.²⁶ In addition, wheat grains (*Triticum aestivum*), exposed to TiO₂ NPs at a dose of 500 mg kg^{−1} revealed an increase in overall amino acid nutrition in an edible portion.³³

2.1.4. Copper oxide (CuO). CuO NPs have gained substantial technological interest due to their high conductivity, cost efficiency, eco-friendliness, ease of availability, and low toxicity. Cu is an essential micronutrient that is incorporated into many proteins and enzymes and plays an important role in plant health and nutrition. Many critical physiological and biological activities in plants are dependent on it, including cellular transportation, mitochondrial respiration, protein trafficking, antioxidative action, and hormone signalling.³⁴

They can be used as growth promoters or even fertilizers because CuO NPs are easily delivered to the shoots and translocated back to the roots *via* the phloem.³⁵

2.1.5. Aluminium oxide (Al₂O₃). Al is one of the most adaptive, cost-effective, and appealing metals. Al₂O₃ comes in a variety of crystalline forms, including α , θ , γ and η , and γ -Al₂O₃; however, γ -Al₂O₃ NPs are the most widely utilized. Because of their extraordinary sorption capacity, high surface features, excellent adsorption capacity, low-temperature changes, and mechanical strength, Al₂O₃ NPs have enormous promise in the agriculture sector.³⁶ Conversely, Al ions are phytotoxic and produce morphological, physiological, and biochemical alterations in plants. However, Al is considered a positive element at low concentrations because it promotes plant development and the uptake of vital components.³⁷ Furthermore, it has also shown antibacterial activities for instance, against Gram-positive (*S. aureus*) and Gram-negative (*E. coli*) at 50, 75, and 100 $\mu\text{g ml}^{-1}$.³⁸

2.1.6. Manganese oxide (MnO₂). MnO₂ NPs are now the subject of research in agriculture because they have significant redox and sorption capabilities that can limit the mobility of harmful metal(loid)s within soils, even under a wide variety of redox circumstances, and minimize contaminant absorption.⁴¹ They have also demonstrated the best vegetative growth characteristics and fruit features, yield, and photosynthetic pigment content. They have a wide range of applications; however, they are most commonly employed as nanofertilizers and antibacterial agents.⁴² They are also beneficial because of their low cost and eco-friendliness, and they have been proven helpful in antibacterial activity due to their surface chemical action. Some of them demonstrated excellent outcomes in developing antioxidant profiles in watermelon seedlings and helped to improve Chl content.⁴³ Furthermore, they displayed a significant reduction in heavy metal arsenic adsorption in rice plants over time and a decrease in translocation factor.⁴⁴

2.2. Biosynthesis of metal oxide nanoparticles

Biosynthesis of MONPs has been of growing importance in nano-biotechnology due to existing various biomolecules in the green/natural sources, which help reduce metal ions to form NPs from bulk metal precursors and stabilize them with biocompatible coatings (Fig. 1(A)). Bioreduction and biosorption are the two main mechanisms followed by biosynthesis.⁴⁵ Natural entities like plants, algae, fungi, and bacteria are used to biosynthesize MONPs. Biomolecules in the sources include alkaloids, saponins, tannins, flavonoids, proteins, carbohydrates, steroids, enzymes, amino acids, cell wall components, and exopolysaccharides.³⁹ (Fig. 1(B)) This biosynthesis approach has a variety of advantages over mechanical and chemical methods, including less expensive, low energy consumption, no need for specialized equipments to carry out the biosynthesis, vast supply of biological sources, and no agglomeration issues because the biomolecules included in the extract stabilize NPs, providing them with biocompatibility^{46,47} (Fig. 1(C)).

For instance, in a recent report, biosynthesis of CeO₂ NPs using *O. europaea* leaves extract was carefully studied.⁴⁰ The HPLC examination of *O. europaea* leaves extract revealed the presence of various bioactive compounds (as shown in Fig. 1 (D)) amongst which oleuropein and oleurosides were the most abundant ones. Moreover, since oleuropein has high reducing power and stronger ionizability, it was predictable that it will play an important part in the capping action. Further, the study hypothesized that as oleuropein rapidly breaks down into elenolic acid and hydroxytyrosol, and since hydroxytyrosol possesses higher polarizing power, the latter acts as a natural capping agent around CeO₂ NPs (Fig. 1(E)).

Furthermore, a recent research comparing the properties and biocompatibility of biosynthesized and chemically synthesized IONPs revealed that green FeO NPs produced using *C. halicacabum* leaf extract had better characteristics due to the capping and stabilizing agents contained in the plant extract.⁴⁹ The study also demonstrated that green IONPs were smaller in size (20.9 nm) than chemically synthesized NPs (42.3 nm), which might be attributable to the diverse range of metabolites found in *C. halicacabum* that aid in reducing and stabilizing Fe ions without agglomeration. Furthermore, the brine shrimp fatality assay revealed that the green IONPs are ecological and demonstrated very less toxicity compared to the chemically synthesized NPs.

2.2.1. Biosynthesis using plant extract. Plants have a special capacity to accumulate some quantity of heavy metals inside their different parts. As a result, biosynthesis techniques based on plant extracts have gained popularity as a simple, productive, cost-effective, and practical way for producing NPs, and also an effective alternative to other synthesis methods. Plants of various varieties can be employed to minimize and stabilize MONPs in biosynthetic methods. Many researchers are working on the biosynthesis process for preparing MONPs using plant extracts to exploit them for different applications.⁵⁰ The biosynthesis process employs complete plants or specific plant components such as roots, leaves, seeds, barks, flowers, fruits, peels, petals, seed husks, and so on. These plant components are high in biomolecules (phytochemicals) such as carbohydrates, amino acids, flavonoids, proteins, saponins, terpenoids, and nitrogenous substances, which act as reducers, redox mediators, and capping/stabilizing agents during NPs formation.⁵¹ For instance, the green synthesis of ZnO and CuO NPs uses an *Achillea nobilis* flowering branch extract.⁵² Likewise, *Catha edulis* leaves extract is used to synthesize CuO NPs.⁵³ Similarly, Fe₂O₃ and TiO₂ NPs can be synthesized using *Koeleria apiculata* and *Kniphofia foliosa* plant extracts.^{54,55}

2.2.2. Biosynthesis using microbes. *Bacteria:* Bacterial species are extensively utilized for numerous biotechnological applications like bioremediation, genetic engineering, and bioleaching. A variety of bacterial species are used for synthesizing metallic NPs due to their ability to reduce metal ions of various metal precursors. Prokaryotic bacteria and actinomycetes are one of the main natural sources which are used for the biosynthesis of MONPs.⁵⁶ Because of their widespread

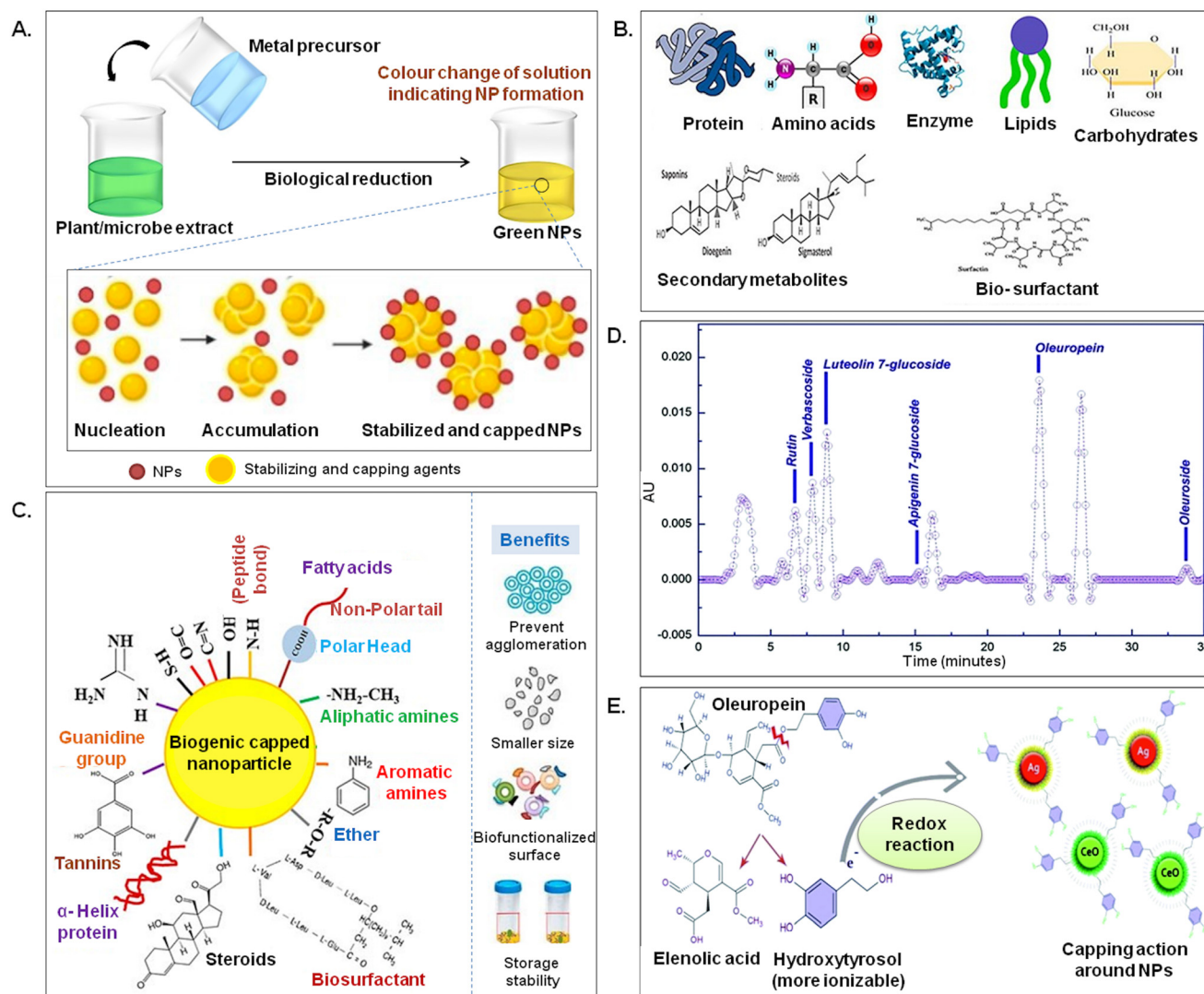


Fig. 1 Biosynthesis of NPs. A. Schematic illustration of various steps involved in biosynthesis using plant/microbes. B. Different biomolecules available in plants/microbes which act as stabilizing/capping agents during biosynthesis. C. Biogenic capped NPs prepared via biosynthesis are less agglomerated, smaller in size, are biofunctionalized (provides functional groups for further biomolecule/chemical attachment) and have higher stability. (B & C adapted from ref. 39 with permission from Frontiers, 2021). D. Analysis of biomolecules present in *O. europaea* leaves extract that can potentially act as stabilizing and capping agents via HPLC examination. E. Schematic illustrating capping action of Oleuropein biomolecule around prepared NPs. (D & E reproduced from ref. 40 with permission from Royal Society of Chemistry, 2017).

prevalence, short doubling time, ability to grow under harsh conditions, and affordable and easy growth media, prokaryotic systems have been intensively researched as a significant paradigm in the field of nanotechnology.⁵⁷ Furthermore, bacteria accumulate and detoxicate metals *via* the activity of the reductase enzyme, hence this trait has been utilized to synthesize MONPs using a reaction precursor. Additionally, enzymes, proteins, peptides, cofactors, and organic compounds operate as stabilizers, capping agents, and reducing agents during the biosynthesis process.⁵⁸ Nonetheless, bacteria-mediated biosynthesis has been regarded as the best method for producing MONPs with a variety of shapes, sizes/distribution, structural frameworks, and physicochemical properties by decreasing metal ions.⁵⁹

Fungi: Fungi-based biosynthesis is also one of the commonly used methods for the synthesis of MONPs with well-defined morphologies. They are good biological agents for the preparation of MONPs, because of the presence of different types of enzymes inside their cells. Generally, fungi-mediated biosynthesis of NPs is more efficient as compared to the synthesis using bacteria. Moreover, fungi have many advantages over other organisms due to the presence of enzymes, proteins, and reducing components on their cell surfaces.^{60,61}

Algae: Algae is one of the ecologically important groups of photosynthetic organisms. They can be unicellular or multicellular according to the conditions they are growing like fresh water, marine water or moist rock surfaces. Algae are also known as bionanofactories because both live or dead dried

biomasses of algae are used for synthesis of MONPs.^{62,63} There are two types of algae to prepare MONPs which include microalgae (unicellular) and macroalgae or seaweeds (multicellular). Algal bodies, like all other green sources, include several biomolecules such as proteins, lipids, carbohydrates, peptides, alkaloids, terpenes, macrolides, cell wall polysaccharides, glycoproteins, and enzymes that play important roles in the reduction, capping, manufacturing, and stability of NPs.⁶⁴ A list of various methods of biosynthesized MONPs, and their source and agricultural applications are provided in Table 2.

3. Plant growth Mechanisms using MONPs

MONPs induce crop growth, biomass production, and enhance grain yield/quality due to different nano-mediated mechanisms mainly at the plant levels *via* photosynthesis, minerals translocation/uptake/assimilation, ROS generation, and genomic alteration. Further, MONPs generate resistance against various abiotic stresses such as drought, salinity, and mineral toxicity in plants through various mechanisms, which are described in the following subsections with the support of reported findings.

3.1. Translocation, uptake, and assimilation of minerals

Absorption and translocation of sufficient levels of crucial mineral elements (Zn, Fe, Al, *etc.*) are vital, especially for early plant growth and development. Many studies have shown that uptake of MONPs has enhanced the assimilation of these minerals in plants.³⁵ Generally, NPs can go through plant tissue either *via* root or above-ground plant parts, including leaves, root junctions, and wounds.⁶⁵ In the case of foliar application (*via* leaves), the NPs enter mainly through the stomata or cuticles. Cuticles serve as major leaf barriers, preventing NPs smaller than 5 nm from entering. NPs bigger than 10 nm, on the other hand, enter through stomata and are transported into the plant's vascular system *via* apoplastic and symplastic pathways.⁶⁶ Smaller NPs (10–50 nm) are more likely to be transferred *via* the cytoplasm of a neighbouring cell (symplastic route), and larger NPs (50–200 nm) are translocated between the cells (apoplastic route).⁴⁸ Further internalization of NPs inside the plant cells/tissues is mediated by Endocytosis (cells taking in substances by engulfing them in a vesicle), pore formation *via* carrier proteins, and plasmodesmata (intercellular channels that cross the plant cell wall and act as cytoplasmic bridges to allow efficient molecular exchange between nearby cells)⁶⁵ as illustrated in Fig. 2.

Several recent studies have demonstrated the impact of MONPs on mineral assimilation through this mechanism. For example, foliar treatment of Fe₂O₃ NPs at 20 mg L⁻¹ on wheat increased Fe concentrations in shoots, roots, and grains by 44, 40, and 69%, respectively.⁶⁷ In another investigation, seed priming with FeO/ZnO NPs raised Fe and Zn concentrations in

shoots by roughly 12 and 24%, respectively, and in roots by approximately 13 and 19%, respectively.⁶⁸ Likewise, foliar treatment of ZnO NPs on wheat increased Zn content in grains by 40–85%, shoots by 10–52%, and roots by 7–60%.⁶⁹ In the same study, when 100 mg kg⁻¹ ZnO NPs were applied to the soil, Zn concentrations increased in shoots, roots, and grains by 58, 68, and 190%, respectively. In another similar investigation, wheat plants treated with TiO₂ NPs had higher concentrations of Fe and Mn in different areas of the plant than the control.⁷⁰ Analogously, the application of CuO-NPs dramatically increased the accumulation of beneficial substances such as vitamin C, lycopene, and flavonoids in tomato fruit.⁷¹ Yet another study reported that applying Al₂O₃ NPs to Lettuce plants increased elemental concentrations of macronutrients dramatically.⁷² Another study corroborated the rise in Al³⁺ or Zn²⁺ concentrations in barley shoots and roots upon treatment with Al₂O₃ and ZnO NPs where the uptake was robust in root tissues.⁷³

3.2. Photosynthesis

Photosynthesis occurs due to two types of reactions: light-dependent reactions and light-independent CO₂ fixation reactions (dark reactions). The plant absorbs energy from sunlight and uses it to build the energy storage molecules Adenosine triphosphate (ATP) and nicotinamide adenine dinucleotide phosphate (NADPH) through light-dependent reactions; then, these end products (ATP/NADPH) are combined with CO₂ to form glucose in the following dark reaction (called Calvin cycle). Because vascular plants can only use 2–4% of the available energy from sunlight,⁷⁶ researchers have been working on developing artificial sunlight-harvesting systems for which MONPs have been proven to be a great candidate due to their inherent light absorbance and intensifying abilities.⁷⁷

Chl is a plant pigment that absorbs light energy and converts it to electrical and chemical energy. Numerous studies have shown that applying MONPs to plants greatly improves Chl a/b and carotenoid (Ctd) content, allowing plants to generate more light-harvesting complexes (LHCIIIs) to absorb more sunlight and promote photosynthesis. LHCIIIs are pigment-protein complexes associated with photosystem I (PSI) and photosystem II (PSII) that harvest photons and use sunlight energy to catalyze the primary photosynthetic reactions that produce high energy compounds. Moreover, due to the photocatalytic properties of MONPs, they can cause redox reactions which lead to the transfer of charge, and hence they can function as electron and proton transporter for PSI, PSII, ATP synthase (catalyst for ATP synthesis), plastoquinone (involved in light-dependent reactions), *etc.* Further MONPs have been shown to enhance the kinetics of the Calvin cycle by increasing the activities of enzymes like rubisco (RuBP), carbonic anhydrase, *etc.*⁷⁸ Fig. 3(I) sums up these different advantageous interactions between MONPs and photosynthetic infrastructure.

Further there has been a lot of research going on for the investigation of the above discussed benefits of MONPs. For instance, a recent study showed that different concentrations

Table 2 Biosynthesis of MONPs using various plants and microbes and their agricultural applications

MONPs	Plant/microbe (biological source/agents used)	Amount of extract & metal precursor (medium) used	Morphology	Agricultural applications	Ref.
Plant extract					
Fe ₂ O ₃	<i>Sageretia Thea</i> (Fine powdered fresh leaves)	30 g	Tetragonal crystalline	AMBs	206–208
		200 ml distilled water (FeSO ₄ ·7H ₂ O)	29 (nm)		
	<i>Matus Pumila</i> (Apple peels)	75 g	Elliptical and spherical	Moderate antibacterial activity	
		300 ml mili-Q water (FeCl ₂ ·H ₂ O; 20 mM, FeCl ₃ ·6H ₂ O; 40 mM, NaOH; 1 M)	50–100 (nm)		
CuO	<i>Ruellia tuberosa</i> (Leaf extract)	5 g	Hexagonal rods	Kills bacterial pathogens	53, 209 and 210
		100 ml deionized water (FeSO ₄ -1 M)	52.78 (nm)		
	<i>Catha edulis</i> leaves extract	10 g	Spherical	Antibacterial activity	
		150 ml boiled water (CuH ₆ N ₂ O ₉)	18.2 (nm)		
ZnO	<i>Sterculia Urens</i> (Gum Karaya)	1 g	Spherical	Antibacterial	211 and 212
		(10 mM CuCl ₂ ·H ₂ O: 0.01 ml)	4.8 (nm)		
	<i>Annona squamosa</i> (Seed Extract)	20 g	39.8 (nm)	Antibacterial activity & is used for the electrocatalytic determination of (H ₂ O ₂)	
		(1 M CuSO ₄ solution)			
TiO ₂	<i>Sedum alfredii</i> Hance Plant	20 g	Hexagonal and pseudo-spherical	Antimicrobial activity	54 and 213
		90% C ₂ H ₅ OH soln. (NaOH, C ₃₂ H ₃₀ ON ₄ Zn (CO ₂ H) ₂)	53.7 (nm)		
	<i>Hibiscus Sadariffa</i> (Leaf)	5 g	Hexagonal wurtzite structure	Phytochemical	
		100 ml deionized water (Zn(NO ₃) ₂ solution)			
Al ₂ O ₃	<i>Kniphofia foliosa</i> (Root Extract)	5 g	Crystallite	Strong antibacterial activity	214 and 215
		200 ml ethanol solution (0.4 M NaOH salt solution)	10.2 (nm)		
	<i>Psidium Guajava</i> (Leaf Extract)	20 g	Spherical	Antibacterial & antioxidant	
		250 ml distilled water (80 mL of TiO(OH) ₂)	32 (nm)		
In ₂ O ₃	<i>Culex quinquefasciatus</i>	—	Spherical	Antipathogens	216
		—	3 (μm)		
Bacteria	Onion (<i>Allium cepa</i>)	—	Hexagonal	Antioxidant	217 and 218
			28 (nm)		
	<i>Aloe barbadensis</i> Miller (Aloe vera Leaves)	35 g	Spherical	Fertilizer	
		100 ml boiled deionized water	5–50 (nm)		
CuO		Dried In ₂ O ₃ precursor			219 and 220
	<i>Streptomyces MHM38</i>	50 ml	Spherical	Antibacterial activity	
		1 mM of CuSO ₄ solution	(200–800 nm)		
	<i>Morganella morganii</i>	Bacteria: 50 ml	Crystalline	Antibacterial Property	
ZnO		Precursor: 0.050 ml	24 (nm)		219 and 220
		CuSO ₄ ·H ₂ O			
	<i>Serratia nematodiphila</i>	Precursor: 0.1 M ZnSO ₄	Spherical	Anitfungal	
		Precursor: 0.2 M Zn (CH ₃ CO ₂) ₂	10–50 (nm)	Photocatalytic activity for agricultural applications and helps in plant growth	
			40 (nm)		

Table 2 (Contd.)

MONPs	Plant/microbe (biological source/agents used)	Amount of extract & metal precursor (medium) used	Morphology	Agricultural applications	Ref.
TiO ₂	<i>Achromobacter</i> sp. M15 (Bacterial Cellulose from agricultural waste)	4 g	10–40 (nm)	Antimicrobial activity & enhances crop production	221 and 222
	Rhizospheric Bacteria	95 ml sterile water Precursor: C ₁₂ H ₂₈ O ₄ Ti soln (5 ml) 10 g 10 ml of 0.85% saline soln. Precursor: [TiO(OH) ₂] soln.	Crystalline 10–17 (nm)	Photocatalyst	
Fungi	<i>Periconium</i>	20 g	Hexagonal wurtzite structure 40 nm	Antibacterial agent	223–225
ZnO	<i>Aspergillus terreus</i>	200 ml deionized water 20 g Zn(NO ₃) ₂ 50 ml 5 mM ZnO	Spherical 8 (nm)	Antibacterial activity	
Fe ₂ O ₃	Manglicolous fungi	Precursor: FeCl ₃ solution	Crystallite structure 2–16 (nm)	Antifungal agent	226 and 227
	<i>Fusarium Oxysporum</i>	Precursor – 2 : 1 molar mixture of K ₃ [Fe(CN) ₆] and K ₄ [Fe(CN) ₆]	Quasi-spherical	Increase crop growth	
TiO ₂	<i>Aspergillus flavus</i> TFR7	Precursor: TiO ₂	20–50 (nm) Spherical 12–15 (nm)	Plant nutrient fertilizer	228 and 229
	<i>Fusarium oxysporum</i>	—	Sphericla 1 to 100 (nm)	Antifungal and antiparasitic agent	
CuO	Endophytic Fungi	1 g 50 m sterilized-potato-dextrose broth. Precursor: 10 mM CuSO ₄	17.65 (nm)	Antimicrobial	230 and 231
	<i>Pseudomonas fluorescens</i> and <i>Trichoderma viride</i>	100 g	Irregular-shaped 40–100 (nm)	Antifungal (fungicide)	
Algae	<i>Gracilaria gracilis</i>	Precursor: 1 g Zn(NO ₃) ₂	Spherical 18–50 (nm)	Crop protection, photocatalysis	232 and 233
ZnO	<i>Sargassum muticum</i>	2 g 100 ml distilled water (Zn (Ac) ₂ ·2H ₂ O) (2 mM)	Hexagonal crystal 30–57 (nm)	Antibacterial	
CuO	<i>Eichhornia crassipes</i>	5 g CuSO ₄ solution	Spherical 32 (nm)	Antifungal agent	234 and 235
	<i>Cystoseira trinodis</i>	10 g 100 ml double distilled Precursor: CuSO ₄ soln. Precursor: FeCl ₃ solution.	Crystallite 6–7.6 (nm)	Antibacterial activity	
Fe ₂ O ₃	<i>Sargassum muticum</i>		Cubic 18 (nm)	Fertilizer	236
	<i>Kappaphycus alvarezzi</i>	10 g	Spherical	Reduction of nitrogen and phosphorus	64 and 237
		100 ml distilled water. FeCl ₃ ·6H ₂ O	14.7 (nm)	Helps in crop growth	

of Fe₃O₄ NPs, significantly increased photosynthetic pigments of *Moringa oleifera* L. as compared with untreated control.⁷⁴ As shown in Fig. 3(II), it is evident that foliar application with Fe₃O₄ NPs at 40 ppm has the most advantageous effect as compared to other treatments, whereas at 60 ppm, the NPs displayed toxicity by reducing the photosynthetic pigments content as compared with the control. In another study, ZnO NPs were biosynthesized using duckweed extract and analysed to be used as fertilizers.⁷⁵ It was observed that the biosynthesized NPs had smaller size (11.7 nm) than the chemically syn-

thesized one (25.5) and further revealed that green ZnO NPs increased chl a and b content in maize at 25, 50 and 100 mg L⁻¹ (as depicted in Fig. 3(III)) but were not much effective at 200 mg L⁻¹. From the above two studies, it can be seen that chemically synthesized Fe₃O₄ NPs showed significant toxicity at 60 ppm whereas biosynthesized ZnO NPs did not show any even at 200 ppm indicating that green MONPs are relatively less toxic. More such recent investigations discussing the impact of MONPs on the rate of photosynthesis in different plants are given in Table 3.

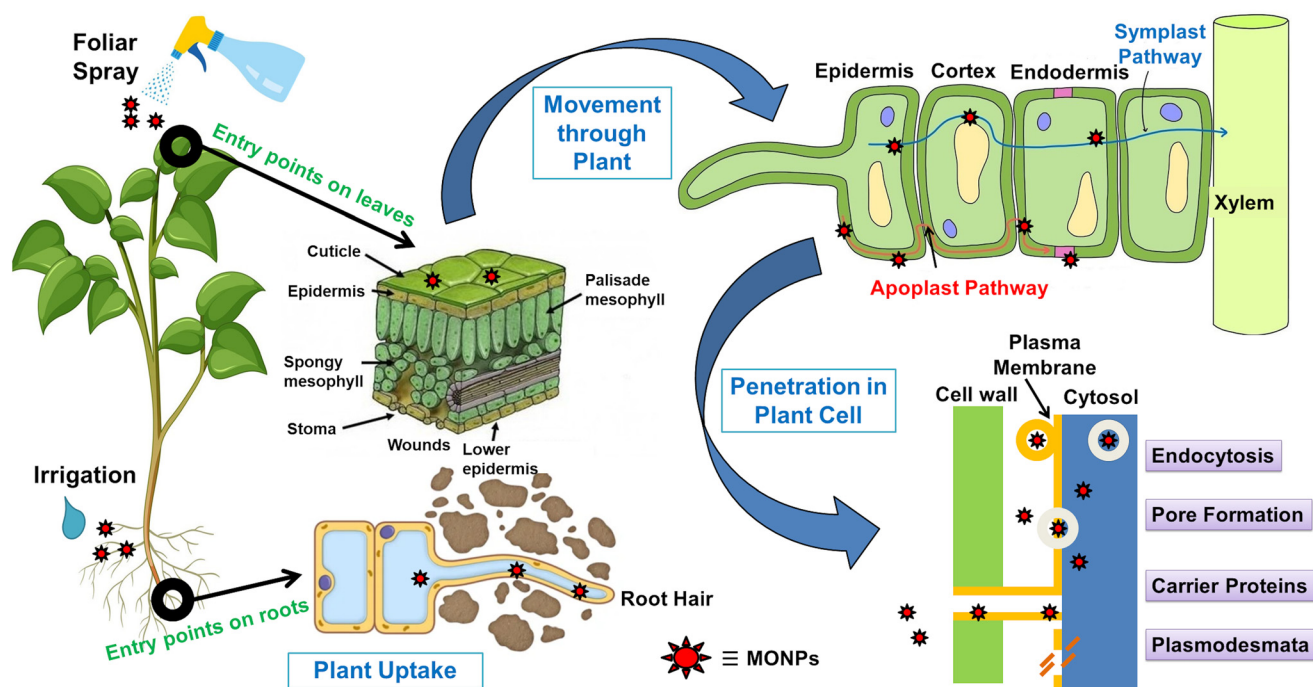


Fig. 2 Schematic representation of NPs absorption routes and translocation mechanisms in different plant sections. To migrate through the plant, NPs can use the apoplastic and/or symplastic pathways. Endocytosis, pore formation mediated by carrier proteins and plasmodesmata have all been hypothesized as mechanisms for NPs internalization inside cells.⁴⁸

3.3. ROS generation and its impact on antioxidative defense system

Oxidative burst is one of the primary plant responses to various biotic and abiotic stressors (like elevated UV intensities and photosynthetically active radiation, drought, extreme temperatures, heavy metal toxicity, pesticides and phytopathogens, *etc.*) which results in the formation of different ROS (like $^1\text{O}_2$, $\text{O}_2^{\cdot-}$, H_2O_2 , $^{\cdot}\text{OH}$), further causing oxidative stress in plants. Oxidative stress damages internal cellular molecules such as nucleic acids, proteins, and lipids, interfering with typical membrane functioning and cell metabolism, causing lipid peroxidation, and finally leading to apoptosis. However, ROS do not just have damaging impact on plants but is also vital for appropriate growth and development of plant cells. Hence it is critical to maintain the minimum level of ROS within the cells. When MONPs are exposed to plants, they produce ROS *via* Fenton and Haber Weiss reactions at locations like mitochondria, chloroplast, *etc.*, as illustrated in Fig. 4(A).^{40,79,80} ROS (generated by MONPs at low concentrations) act as signalling molecules, activating the plant antioxidant defense mechanism, which includes both enzymatic agents like superoxide dismutase (SOD), catalase (CAT), ascorbic peroxidase (APX), guaiacol peroxidase (GPX), dehydroascorbate reductase (DHAR), and glutathione reductase (GR) and non-enzymatic antioxidants like ascorbate, Phenolics, Ctd, *etc.*⁸¹ Moreover, MONPs have been displayed to enhance phenolics and peroxidase (POD) activities. Phenolics are vital components of plant antioxidant systems that enhance plant resistance to numerous environmental stresses by triggering hypersensitive responses and cell wall lignifica-

tions, strengthening the plant vascular body. POD is also involved in lignin production to improve plant defense against many diseases by reinforcing plant cell-wall barriers.⁷¹ These enhanced enzymic activities with MONPs can also be attributed to metallic ions which act as activated co-factors for enzyme-substrate bridging. In addition to activating antioxidant defense system, it is suggested that MONPs on itself too can scavenge excess ROS and reduce oxidative stress due to their variable oxidation states and high redox potential. For example, a comparative study between CeO_2 and Ag NPs towards their impact on *L. esculentum* revealed that CeO_2 NPs demonstrated excellent ROS scavenging activity unlike Ag NPs (as shown in Fig. 4(B)) due to the possession of diverse oxidation states of Ce^{3+} (Ce_2O_3) and Ce^{4+} (CeO_2).⁴⁰ Typically, Ce is available in CeO_2 form at CeO_2 -NPs' surface, with inadequate oxygen and valency of Ce^{3+} rather than Ce^{4+} , which can provide the MONPs with high antioxidant capability. Furthermore, in the dissolution stage, the interconversion of Ce^{3+} into Ce^{4+} can create a favourable surrounding for antioxidant enzyme mimicking function. Besides, the ROS produced by MONPs at low concentrations act as messengers to several hormones like auxin, gibberellic acid, salicylic acid, *etc.* (as shown in Fig. 4(A)) that mediate root gravitropism (resulting in vertical root growth), seed germination and hypersensitive responses that ultimately result in enhanced plant growth and stress tolerance.

Many recent studies have established the above-mentioned ROS-induced plant growth mechanism. For instance, Fe_2O_3 NPs boosted antioxidant defense and activity of enzymes like GPX, APX, GR, and CAT in root and shoot of *Dracocephalum*

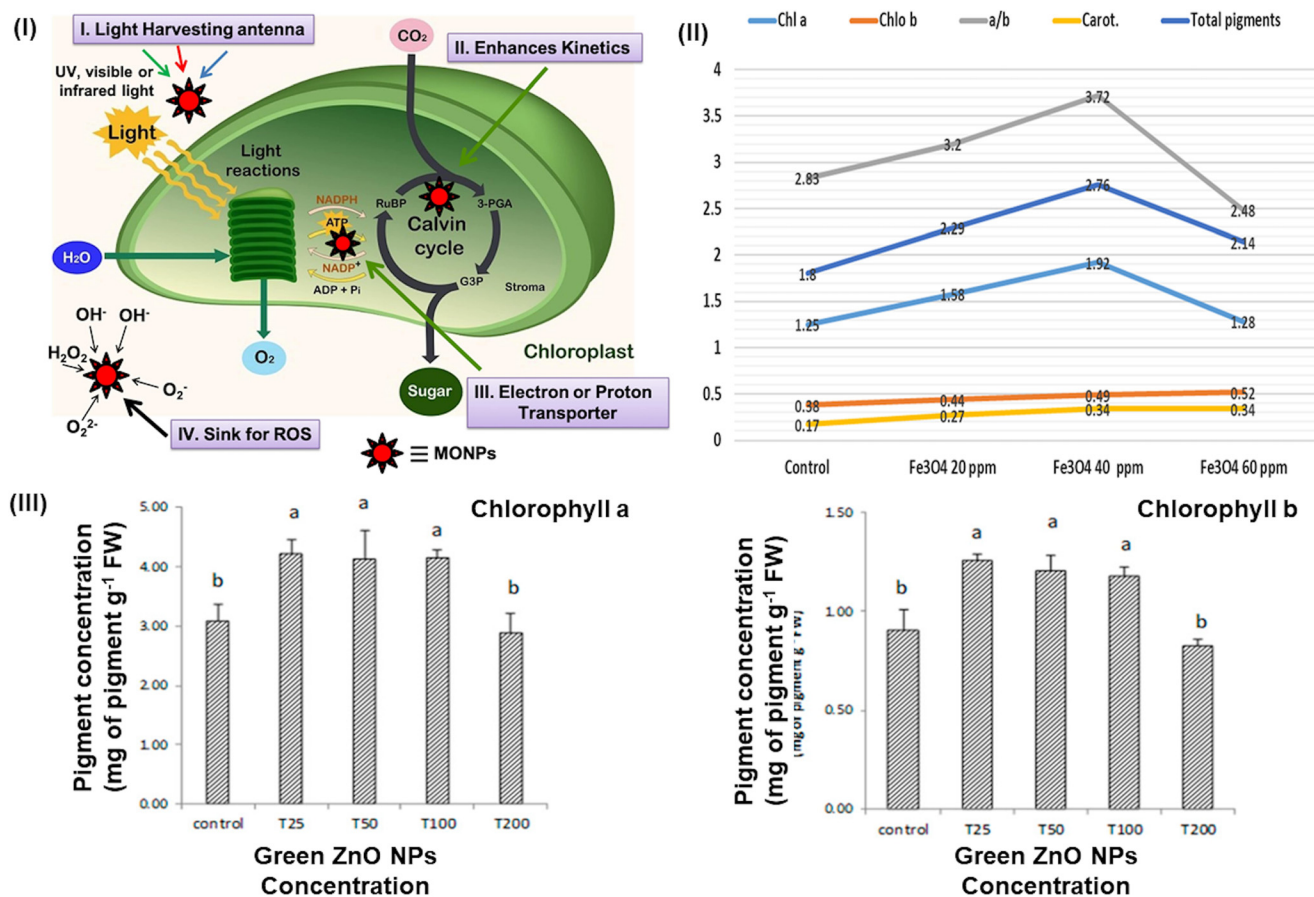


Fig. 3 (I) Potential advantageous interactions between NPs and photosynthetic infrastructure [3-PGA: 3-phosphoglyceric acid; G3P: glyceraldehyde 3-phosphate]. (II) Effects of foliar treatment of Fe₃O₄ NPs on the photosynthetic pigment concentration of *Moringa oleifera* L. (reproduced from ref. 74 with permission from Springer, 2021). (III) Chl a and b content measured 14 days following treatments with green ZnO-NPs (25, 50, 100, and 200 mg L⁻¹) and comparison with the control (different letters in the figure show statistically significant changes for $p < 0.05$ across treatments) (reproduced from ref. 75 with permission from MDPI, 2021).

moldavica L. plant under salinity stress.⁸⁴ Likewise, IONPs significantly enhanced stress tolerance against Cd contamination in soil by boosting SOD and POD activities in shoots of wheat plants.⁶⁷ SOD and POD activity rose by 47 and 50% when wheat leaves were treated with IONPs and ZnO NPs, respectively. In the same study, when wheat plants were treated with ZnO NPs, SOD activity increased by 40% and 46%, and POD activity increased by 46% and 54% in foliar and soil application, respectively.⁶⁹ In a similar report, ZnO NPs were examined on healthy tomato plants and the ones infected with *Tomato mosaic virus*.⁸⁰ It was revealed that ZnO NPs significantly boosted activities of various ROS scavenging enzymes like APX and CAT for both healthy and infected plants (Fig. 4 (C)). Another investigation demonstrated that TiO₂ NPs increased the activities of various enzymic antioxidants, including SOD (75%), CAT (123%), POD (29%), and APX (57%), when applied to broad bean plants.⁸⁵ In a similar study, at 400 mg kg⁻¹ exposure of TiO₂ NPs, the activity of SOD, POD, CAT, and APX in wheat was increased by 111%, 61%, 66%, and 78%, respectively.⁷⁰ However, when the plant was exposed to the same TiO₂ NPs at 600 mg kg⁻¹, SOD, POD,

CAT, and APX activities were lowered. Another study found that enzymic antioxidant activities were consistently enhanced after treatment of CuO NPs on a tomato plant in the concentration range of 5–300 g ml⁻¹. On the other hand, treatments with CuO NPs in the range 300–350 g ml⁻¹, a declining tendency in POD activity, suggesting the toxicity of MONPs at high concentrations.⁷¹ In another investigation, the effects of Al₂O₃ NPs on *Triticum aestivum* plant were studied. The results showed that the concentration of H₂O₂ amplified depending on NPs' dosage, thereby, SOD enzyme activity also increased.⁸⁶ Similarly, on exposure to Al₂O₃ NPs, H₂O₂ content in the roots of *N. arvensis* plant increased at concentrations of 50 and 100 mg L⁻¹, however decreased significantly at 1000 and 2500 mg L⁻¹ concentrations of the NPs, yet again implying toxicity at higher concentrations by scavenging excessive amounts of ROS via antioxidant enzymes.⁸⁷

3.4. Gene expression

As discussed above, various abiotic stressors affect plants throughout their life cycle. Thus, enhancing their resistance to environmental stress at various phases is critical to guarantee

Table 3 Impact of various MONPs on the rate of photosynthesis in different plants at their different concentrations

MONPs used	Morphologies	Type of plants	Mode of applications/ concentrations	Impacts on photosynthesis	Ref.
Fe ₂ O ₃	10 nm	Wheat	Foliar application	• Enhanced Chl a, Chl b and Ctd concentrations by 70%, 139%, and 119%, respectively	67, 238 and 239
	Crystalline		20 mg l ⁻¹		
	Spherical		Soil application	• Enhanced Chl a, Chl b and Ctd concentrations by 53%, 137%, and 111%, respectively	
			20 mg l ⁻¹		
			Foliar/soil application	• Increased photosynthetic rate – 102%, and 90%	
	4–9 nm	<i>Lemna Gibba</i>	19.97 mg l ⁻¹	• Increased transpiration rate linearly	
	Spherical		—	• Decreased Chl content by 51%	
	50–100 nm	<i>Chlorella Vulgaris</i>	0, 12.5, 25, 50, 100, 200		
			400 mg l ⁻¹		
			—	• Increased ROS generation by 2-fold	
				• Decreased cell division rate by 47.8%	
ZnO	20–30 nm	Wheat	12.5 to 400 mg l ⁻¹	• Decreased cell viability significantly	68, 69, 91 and 240
			99.88 mg l ⁻¹	• Increased ROS generation significantly	
			Foliar application	• Chl a increased by 55%, Chl b – 133%, carotenoid 112%	
			25, 50, 75, and 100 mg l ⁻¹	• Photosynthetic rate, transpiration rate, stomatal conductance – increased with NPs' concentration	
				• Photosynthetic rate increased by 25, 61, 90, and 102% respectively	
	2 nm		Soil application	• The photosynthetic rate increased by 18, 46, 77, and 89%	
	20–45 nm	Tomato	25, 50, 75, and 100 mg l ⁻¹		
	—		800 mg l ⁻¹		
			20 mg l ⁻¹	• Reductions in Chl a and b content	
				• Chl a/b ratio increased	
				• Increase in effective photosystem II (PSII) operating efficiency parameter and the plastoquinol redox status parameter.	
TiO ₂	20–25 nm	Broad Bean	0.01% (5 mg l ⁻¹)	• Increased Chl b level by 24%	70, 85 and 241
	—		0.02% (5 mg l ⁻¹)	• Growth and photosynthetic pigment accumulation reduced slightly increased	
	—		0.03% (5 mg l ⁻¹)	• Not much change	
	20–30 nm	Wheat	600 mg l ⁻¹	• Chl a increased by 51%, Chl b by 31%, and Ctd by 31%	
	14 nm	<i>Vetiveria zizanioides</i>	(30, 60, 90, 120, and 150) mg l ⁻¹ NPs' foliar application	• Chl level increased by 27.2% & maximum photochemical efficiency of PSII by 23.5%	
				• The content and yield of essential oil (EO) increased by 23.6 and 55.1%	
				• The khusimol (main active constituent of EO) content and yield were improved by 24.5 and 93.2%	
CuO	Crystallite	Tomato	300 mg l ⁻¹	• Chl and carotenoid contents increased by 90% and 70%, respectively	71, 242 and 243
	10–30 nm	Cucumber	50, 100, 100 mg l ⁻¹	• Significant decrease in Chl a and b contents as the concentration of NPs increased	
	—	Mustard	0, 2, 4, 8, and 16 (mg l ⁻¹)	• Increased the growth and enhanced the photosynthetic efficiency of plants	
Al ₂ O ₃	60 nm	<i>Arabidopsis thaliana</i>	(0.01 mg l ⁻¹), (0.1 mg l ⁻¹), (1 mg l ⁻¹), (10 mg l ⁻¹)	• Chl fluorescence parameters not significantly affected; PSII was not injured	73, 86 and 116
	40 nm	Barley seedlings	50, 250, 500 mg l ⁻¹	• No difference in Chl content	
	Spherical				
	13 nm	<i>Triticum aestivum</i>	5, 25 and 50 (mg l ⁻¹)	• Reduced Chl and Ctd	
				• Chloroplast membrane integrity disrupted; decrease of photosynthetic activity	
				• Inhibition of plant growth and development	

ing normal plant development.² Several studies have shown that after the internalization of MONPs in plant cells, the expression of stress-responsive genes is enhanced to promote plant growth further, as depicted in Fig. 5(I). MicroRNAs (miRNAs) are short, single-stranded noncoding RNA sequences found in all eukaryotic species. MicroRNA is critical in the control of post – transcriptional gene expression because it

attaches to the target messenger RNA (mRNA) concurrently and cleaves or restricts its translation, hence suppressing gene transcription. Plants' miRNA molecules play a range of biochemical tasks, including growth and development regulation and responses to stress. Moreover, some plant's miRNAs have the ability to modulate target gene expression, allowing the plant to adapt amidst environmental changes.^{88,89}

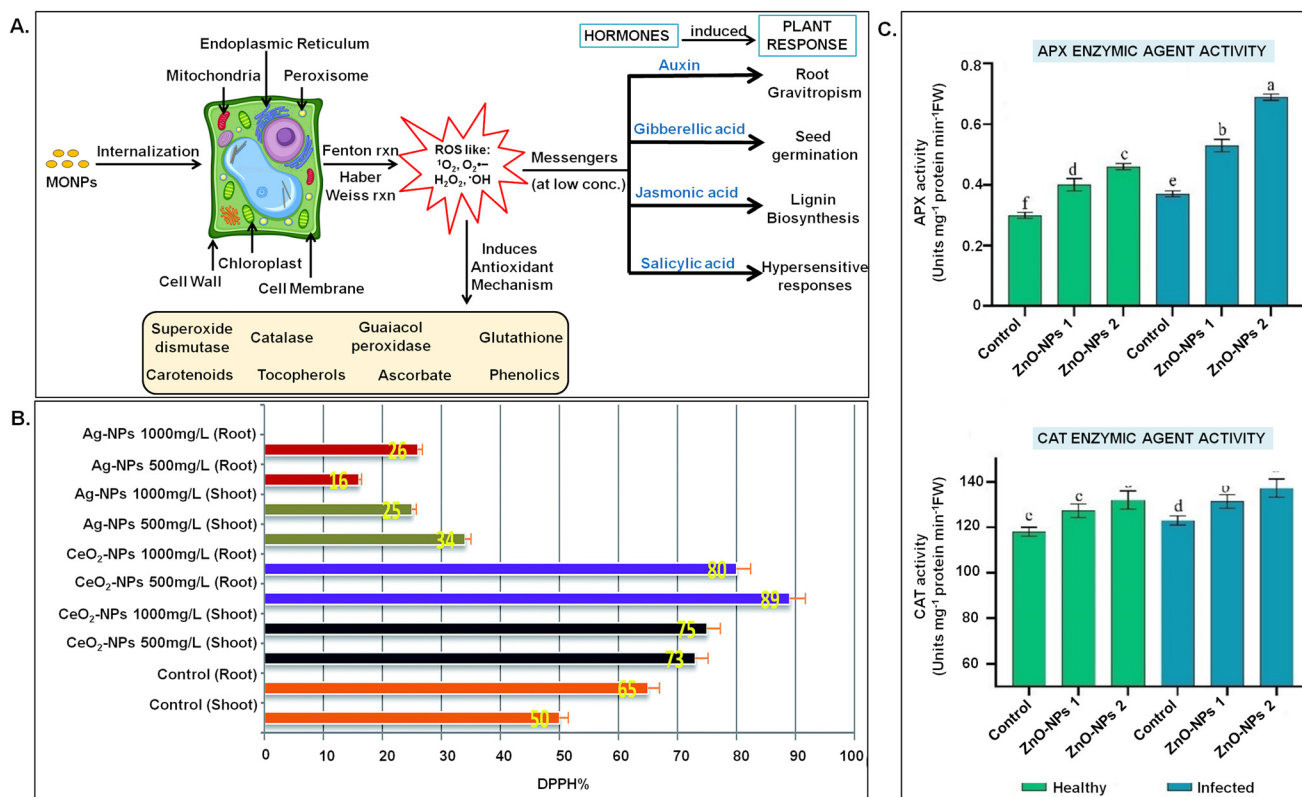


Fig. 4 A. ROS generation due to the application of MONPs at low concentrations and their effect on plant growth via activation of the antioxidant defence system and induction of various hormonal responses.⁷⁹ B. Examination for ROS scavenging with NPs treatment on *L. esculentum* (reproduced from ref. 40 with permission from Royal Society of Chemistry, 2017). C. Impact of ZnO-NPs 1 (50 mg L⁻¹) and ZnO-NPs 2 (100 mg L⁻¹) on APX and CAT enzyme activities of healthy tomato plant and plant infected with Tomato mosaic virus (reproduced from ref. 80 with permission from MDPI, 2021).

Many recent studies have demonstrated this mechanism; for instance, to investigate the mechanism and regulation of ZnO NPs by *Brassica* plants, researchers examined the expression of two genes: BjMTP1 (metal tolerant protein) and BjCET2 (cation efflux transporter).⁸² The study found that at 20 $\mu\text{g ml}^{-1}$ concentration, BjMTP1 unregulated, but BjCET2 showed no major change (as shown in Fig. 5(II)), implying that ZnO NPs develop plant tolerance to metal ions up to 20 $\mu\text{g ml}^{-1}$. But, BjMTP1 downregulation and BjCET2 overexpression at 30 $\mu\text{g ml}^{-1}$ show that concentrations beyond 20 $\mu\text{g ml}^{-1}$ are hazardous to plants. In a similar investigation, miR156 and miR159 (miRNAs) expression levels were analysed in barley seedlings being exposed to ZnO NPs.⁸³ The miRNAs miR156 and miR159 are implicated in a variety of plant stress responses, including drought, hypoxia, fungal infections, and NPs. The qRT-PCR technique was used to assess miRNA expression levels, and the results showed that ZnO NPs' treatment regulated the expression of all miRNAs (miR156a, miR159a, and miR159c) depending on the dosage (Fig. 5(III)). Expressions of MiR156a and miR159a were considerably increased at all dosages; though, when compared to controls, 2 mg L⁻¹ and 4 mg L⁻¹ NPs' exposure considerably downregulated miR159c expression. This demonstrates how MONPs can aid in regulating miRNAs to control stress responsive genes in plants. In another study,

upon exposure to CuO NPs, various differentially expressed genes in the roots and leaves of Zucchini were uniquely modulated, due to which mitochondrial and chloroplast functions were significantly enhanced.⁹⁰ Another study found that foliar application with ZnO NPs boosted the expression of genes of antioxidative enzymes, transporters, and carbon/nitrogen metabolism enzymes, resulting in higher levels of antioxidants, sugars, and amino acids in Fe-deficient tomato plants.⁹¹ In a similar study, aquaporins (channel proteins) and transcription factor genes were differentially regulated in leaves/roots of barley in response to Al₂O₃ and ZnO NPs, which enhanced tolerance against various abiotic stresses.⁷³ Another investigation analyzed the effect of Fe₂O₃, CuO, and ZnO NPs on soybean plants amidst drought-induced conditions. This study reported an upregulation of gene expression depending on the type of gene, tissue and NPs, which helped enhance the drought tolerance of MONPs-treated soybean plants.⁹²

4. Functionalization of MONPs and agricultural waste management

Controlling the size, shape, stability, and dispersibility of MONPs in preferred solvents is a challenging concern as they

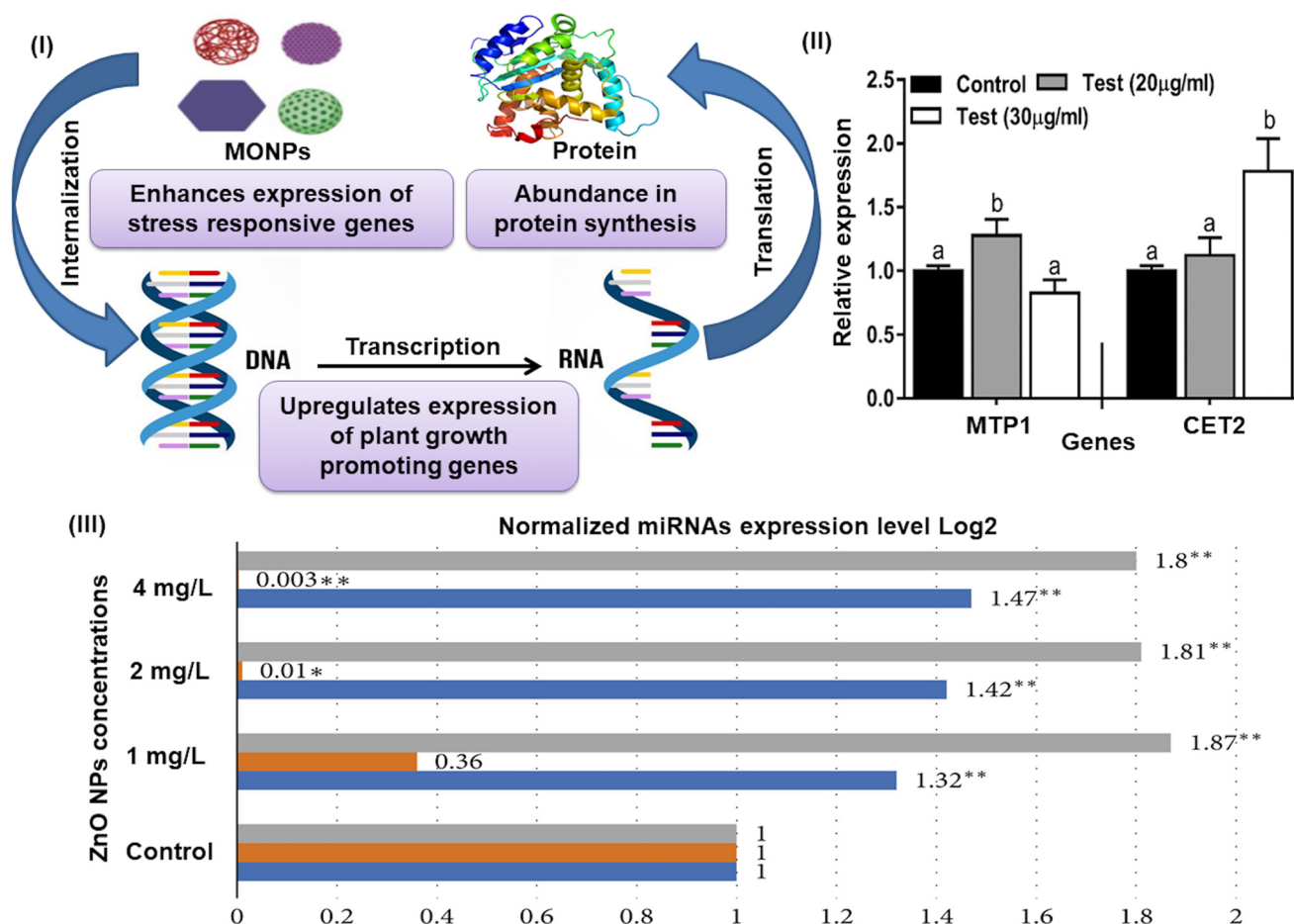


Fig. 5 (I) Representing improvement of stress responsive and growth promoting genes by upregulating expression of microRNAs (i.e. short non-coding RNAs) with MONPs treatment. This further leads to protein abundance and enhanced plant growth.⁴⁸ (II) qRT-PCR analysis of MTP1 and CET2 gene expression under green ZnO NPs exposure at 20 µg ml⁻¹ and 30 µg ml⁻¹ on *Brassica juncea* crop plant (letters a and b designate a considerable difference between the treatments at $p < 0.05$) (reproduced from ref. 82 with permission from Scientific Reports, 2020). (III) miR156a, miR159a, and miR159c expression levels in barley (*H. vulgare* L.) plants subjected to varied concentrations of ZnO nanoparticles in control and experimental groups (values represent the mean of three replicates with standard deviation). * considerable difference from control ($p < 0.05$); ** considerable difference from control ($p < 0.01$) (reproduced from ref. 83 with permission Hindawi, 2020).

tend to agglomerate to reduce their high surface energy. Agglomerated MONPs act very much like their bulk counterparts, thereby undermining the intrinsic advantages of nanoscale materials. The increased size due to agglomeration can alone prohibit MONPs from being taken up by plants and translocated *via* the vasculature in their intact state, thus lowering their biocompatibility and uptake efficiency.^{93–95} As discussed above, biosynthesis can offer a solution to these challenges to some extent due to the existence of natural stabilizing/capping agents (phytochemicals) in plant/microbial extracts, however, biosynthesis alone cannot overcome these issues efficiently. Therefore, it is very important to implement various functionalization pathways to improve the stability, bioavailability, and uptake efficiency of MONPs by altering their physiochemical characteristics through surface engineering/modifications.⁹⁶ These pathways include grafting or coating MONPs with small molecules such as small organic molecules/surfactants, encapsulating within polymers as well

as inorganic shells such as silica, gold and other metal, metal oxide, or metal sulphide layers.⁹⁷ In practice, it is worth noting that in many circumstances, the surface coatings not only stabilize MONPs, but may also provide functional groups for further conjugation/functionalization. Furthermore, functionalized NPs have been widely used in agriculture as nanocarriers for the encapsulation and trapping of agrochemicals such as fertilisers, insecticides, herbicides, plant growth promoters, and other active compounds. These nanocarriers are capable to provide regulated and gradual release of agrochemicals, allowing for slow absorption of active components.^{97,98} Besides, the nanocarriers can be further functionalized with DNA, enzymes, antibodies *etc.* to enable targeted delivery of active ingredients in agriculture. As a result, agrochemicals' efficacy can be boosted considerably as they can securely attach to the plant surface, limiting the quantity of agrochemical waste that reaches the environment.⁹⁹ Nowadays, MONPs are gaining significant attention from researchers for having

great potential to replace conventional agrochemicals (as discussed in the following sections). Numerous recent studies demonstrated the use of functionalized MONPs for sustainable agricultural applications. One of the most widely explored applications of functionalized MONPs is agricultural waste management due to their enhanced adsorption properties.

The widespread usage of conventional agrochemicals (such as pesticides, herbicides, antibiotics and organic chemicals) in agriculture has raised serious environmental issues because of their non-biodegradability and non-selective toxicity/accumulation in the environment, including water and soil. Furthermore, hazardous metal accumulations such as Pb^{2+} , Ni^{2+} , Zn^{2+} , Cr^{3+} , Cu^{2+} , and Cd^{2+} have been producing major environmental and health concerns and hence it is important to find sustainable technologies for agricultural waste management and MONPs especially functionalized IONPs have demonstrated great results in this regard. For instance, in a recent study, amine-functionalized magnetic IONPs (MIONPs-NH_2) were successfully synthesized *via* a simple co-precipitation method using (3-aminopropyl)triethoxysilane (APTES) for amine-functionalization, which exhibited excellent adsorbent properties for cationic-type heavy metals.⁹³ It was revealed that MIONPs-NH_2 could remove metal cations rapidly depending on the complexity of amino groups on their surface. The study suggested the application of MIONPs-NH_2 in wastewater treatments involving multiple heavy metal cations. In a similar report, IONPs clusters were incorporated into amino-functionalized SBA-15 in order to obtain a magnetically recoverable adsorbent.⁹⁴ The results revealed quantitative removal of glyphosate contamination in real water matrices along with complete regeneration of the sorbent through diluted NaOH solution. In a similar study, functionalized lignin-based hybrid magnetic NPs were prepared by a facile method using epichlorohydrin as a cross-linker between amino-functionalized magnetic NPs and carboxymethylated lignin.⁹⁵ The hybrid NPs showed ultrafast adsorption capabilities for Pb^{2+} and Cu^{2+} removal. In another recent report, IONPs were biosynthesized using black tea and were further functionalized with 1-butyl-3-methylimidazolium bromide.¹⁰⁰ These functionalized IONPs exhibited excellent adsorption properties and could successfully remove cyanazine from water, a herbicide applied for broad leaf weeds and yearly grasses in the growing fields of bananas, pineapple, sugarcane. Further, Fe_3O_4 -graphene oxide nanocomposite was functionalized with beta-cyclodextrin which exhibited high adsorptions for four kinds of fungicides.¹⁰¹ In a similar study, the self-assembly of Fe_3O_4 NPs on the surfaces of carbon nanospheres resulted in a unique magnetic adsorbent that was employed in the removal of eight triazole fungicides from ambient water sources. The results demonstrated a high extraction percentage (above 80%) of the prepared nanoabsorbent in all cases.¹⁰² Analogously, a Zr-based magnetic metal-organic framework was developed by immobilizing UiO-66 onto $\text{Fe}_3\text{O}_4/\text{SiO}_2$ particles by a proficient single-pot solvothermal method and was used for adsorption of triclosan (TCS) and triclocarban (TCC) fungicides from water by magnetic solid phase separation.¹⁰³ It was revealed that high saturated adsorption capacities were attained

for TCS and TCC removal and the process was convenient, rapid and eco-friendly and can be recycled more than 11 times. Further another investigation revealed that functionalized magnetic NPs can be used for efficient adsorption of agricultural wastewater contaminated with antibiotics, pesticides and toxic metals.¹⁰⁴ In summary, functionalized MONPs can be successfully implemented for agricultural waste management. Functionalization of MONPs to optimize their nutrient delivery, antimicrobial and pesticidal activities, and sensing abilities have been discussed in the following sections.

5. Application of MONPs in sustainable agriculture

The usage of MONPs-based nanofertilizers (NFZs), nanopesticides (NPDs), antimicrobial agents (AMBs), and nanobiosensors (NBSs) in agriculture is a rapidly expanding research field because of their unique properties and associated mechanisms (as discussed in the above sections). Compared to conventional agrochemicals, studies of MONPs-based NFZs/NPDs/NBSs/AMBs have indicated accelerated germination, increased plant resistance to biotic/abiotic stresses, favoured effective nutrient management, and increased plant development while decreasing negative environmental impacts. In the following sections, sustainable agricultural applications of biosynthesized MONPs (including functionalized MONPs) have been discussed in detail in support of recent reports.

5.1. Nanofertilizers (NFZs)

NFZs have an inimitable potential to supply plant nutrients with a high absorption rate and thereby increasing their usage efficiency with minimal losses. Hence, NFZs are a superior alternative for facilitating nutrients uptake in plants to improve various growth parameters such as shoot length (SL), root length (RL), shoot dry weight (SDW), root dry weight (RDW), shoot fresh weight (SFW), root fresh weight (RFW) and leaf area index (LAI).¹⁰⁵ Moreover, compared with conventional fertilizers, the unique properties of MONPs give them an advantage of enhanced crop productivity *via* site-specific targeted delivery and controlled release of nutrients besides providing plant protection.¹⁰⁶ Although several research works displayed exponential plant growth using biosynthesized MONPs-based NFZs at low concentrations, it causes a nutritional imbalance with high bioaccumulation of MONPs, leading to a decrease in crop growth.⁶⁵ Overall, NFZs have more advantages over conventional fertilizers because they (i) boost soil fertility, crop yield, and crop quality; (ii) are nontoxic and less damaging to the environment and humans; (iii) reduce the cost of production and optimize profit.¹⁰⁷

MONPs such as ZnO NPs, CuO NPs, $\text{Fe}_2\text{O}_3/\text{Fe}_3\text{O}_4$ NPs, and MnO NPs are commonly used to pack micronutrient nanofertilizers comprising Zn, Cu, Fe, and Mn, respectively.¹⁰⁸ For instance, SL/RL/SDW/RDW/SFW/RFW/LAI of *Dracocephalum moldavica* L. plant was increased with Fe_2O_3 NPs treatment at 90 ppm under salinity stress.⁸⁴ It was also shown that the con-

centration of phenolic compounds and flavonoids increased due to the favourable effect of Fe_2O_3 NPs on salt stress resistance, particularly in the roots. These findings suggest that Fe_2O_3 NPs enhanced plant nutritional requirements, minimising the negative impacts of salt stress. Another study examined the effect of foliar application of Fe_2O_3 NPs on the growth and physiological development of soybean seedlings. The positive benefits of soybean were shown to be size and concentration dependent. Foliar application of the smallest size – Fe_2O_3 NPs ($\text{S-Fe}_2\text{O}_3$ NPs, 4–15 nm, 30 mg L^{-1}) resulted in the highest growth promotion, increasing shoot and nodule biomass by 55.4 and 99.0%, respectively, which is 2.0- and 2.6-fold higher than the commercially available Fe fertilizer (EDTA-Fe) with equivalent molar Fe. Besides, $\text{S-Fe}_2\text{O}_3$ NPs increased soybean nitrogen fixation by 13.2% more than EDTA-Fe. Furthermore, $\text{S-Fe}_2\text{O}_3$ NMs boosted carbon absorption in nodules, supplying greater energy for nitrogen fixation, according to transcriptome and metabolomic investigations.¹⁰⁹ Likewise, it was observed that treatment to Fe_3O_4 NPs boosted symbiotic nitrogen fixation and increased total nitrogen content in the shoot (26.5–50.2%) and root (24.1–48.2%) of exposed common bean plants as compared to controls.¹¹⁰

Furthermore, several studies have suggested that MONPs might reduce hazardous metal absorption and toxicity in plants by improving plant growth and photosynthesis, and ZnO NPs have effectively proved this in recent reports. For example, the SL, RL, and SDW of the wheat plant were increased by 53%, 46%, and 74%, respectively, with ZnO NPs' exposure at 100 mg L^{-1} by improving the Chl content amidst Cd stress.⁶⁸ Cadmium toxicity may result in a lack of vital plant nutrients due to competition from Cd and other divalent ions during plant absorption. In a similar study, it was revealed that ZnO NPs increased the Zn concentrations and reduced the Cd concentrations in shoot, root, and grains of wheat compared to control. Moreover, treatment of ZnO NPs on the wheat plant at 25, 50, 75, and 100 mg kg^{-1} increased SL by 21, 35, 50, and 55% in the foliar application and by 10, 23, 33, and 43% in soil application respectively.⁶⁹ In another study, green ZnO and MgO NPs were synthesized using Aloe vera leaf extract and used to improve seedling development of *Vigna radiata* (mung bean).¹¹² The results demonstrated that RL, SL, fresh biomass and dry biomass; seed germination rate; and chl content in mung beans significantly increased as compared to the control on treatment with both ZnO and MgO NPs. It was also discovered that ZnO NPs could access the seed coat more easily than MgO NPs or the control.

Further, applications of CuO and MnO NPs in agricultural systems have also demonstrated promising results in terms of crop yield enhancement when compared to their bulk equivalents CuSO_4 and MnSO_4 .¹¹³ CuO NPs, for example, were biosynthesized using green tea extract (which acts as a reducing and capping agent). The study also discovered that green tea produced CuO NPs at concentrations ranging from 0.2 to 20 g mL^{-1} are non-phytotoxic and may promote lettuce radicle development. But, at higher concentrations (40 to 300 g mL^{-1}), these NPs inhibited radicle development in lettuce, indicating

that CuO NPs can be used as nutrient elements; however, depending on their dosage, size, surface characteristics, delivery mode, and exposure period, they can be detrimental.¹¹⁴ In a similar report, the number of leaves, roots formed per explants, and RL were considerably increased for banana plants treated with biosynthesized CuO NPs using fungal metabolites at 9 mg L^{-1} .¹¹⁵ It was further concluded that CuO NPs could improve growth parameters and rooting of banana plants higher than bulk CuSO_4 .

Besides essential micronutrients, MONPs such as TiO_2 can assist plants in nutrient absorption without contributing any nutritional element themselves. For example, the application of 0.01% TiO_2 NPs enhanced plant growth, which was manifested in the form of increased SL, LAI, and RDW by 48%, 36%, and 44%, respectively, when treated on broad bean plants under salinity stress.⁸⁵ In another recent study, it was evident that the application of TiO_2 NPs (at 200 and 400 mg kg^{-1}) increased spike length, SL, and SDW in wheat plants. However, the values of subjected parameters decreased at a high concentration of NPs (600 mg kg^{-1}) compared to lower concentrations.⁷⁰ Al_2O_3 NPs are one of the most prevalent nanoparticles generated in the industrial and commercial sectors, and they are now being investigated for their applications in agriculture. A recent study showed that Al_2O_3 NPs at 100 mg L^{-1} stimulated the growth of *N. arvensis* plant but exerted toxicity at higher concentrations.⁸⁷ In another study, Al_2O_3 NPs treatment increased RDW and RL of *Arabidopsis thaliana* plant by 47.07% and 38.53%, respectively, and even enhanced the thickness and length of its root hairs.¹¹⁶ Likewise, the exposure of 0.4 mg mL^{-1} of Al_2O_3 NPs produced a 17.9% increase in biomass. However, the exposure of 1 and 2 mg mL^{-1} of Al_2O_3 NPs revealed a 10.4% and 17.9% decrease in biomass compared to the control.⁷² Hence, it is confirmed that MONPs can only improve plant growth and morphology when administered at low concentrations.

Furthermore, MONPs can be functionalized to improve their efficiency and control their release. For instance, in a recent study, a smart nanocomposite of chitosan/alginate nanoparticles loaded with CuO was biosynthesized and analysed to be used as smart NFZs.¹¹¹ Fig. 6(I) shows that the final structure comprised of a PEC coating composed of CS, STTP, and ALG wrapped around CuO NPs. The function of polymeric coating in the release of Cu ions from CuO/PEC NPs was investigated over a 22-day period using release tests as shown in Fig. 6(IIa). The red line shows that the CuO NPs release Cu ions rapidly in the solution over the first two days, reaching a percentage release of about 90% on day 22. In the presence of the polymeric covering, however, the profile (black line) shows substantially slower Cu ion release. As shown in Fig. 6(IIb), two primary processes may be suggested to explain the Cu release: Cu diffusion out of the polymeric nanoparticles, driven by the concentration gradient, and polymer swelling and breakdown, which occurred over many days. Further as shown in Fig. 6(III), all of the germination metrics exhibited the greatest results when exposed to CuO/PEC nanocomposites at 10 ppm, while in certain cases, CuO/PEC showed the poorest results at 100 ppm, indicating its potential toxicity at

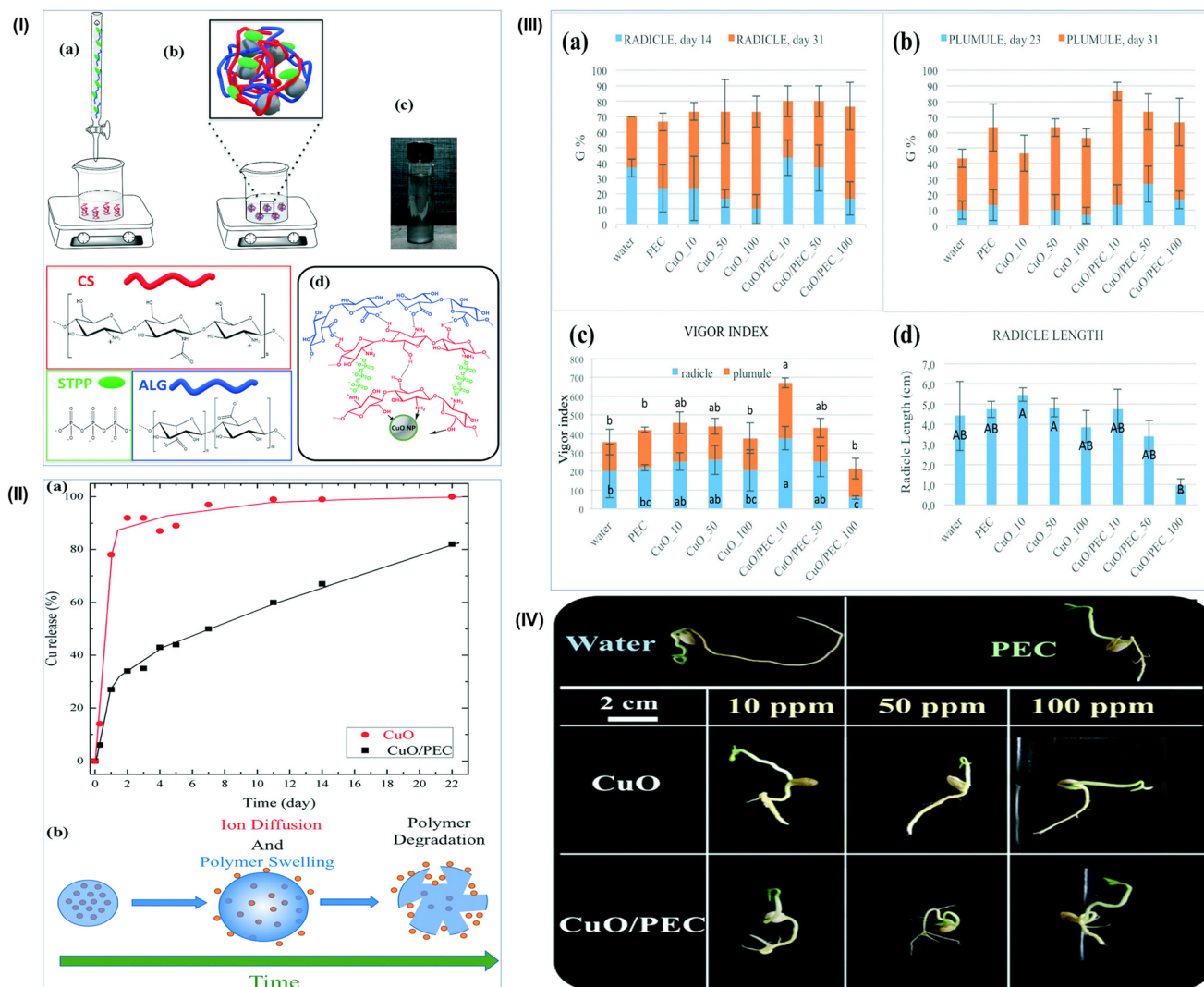


Fig. 6 (I) (a) and (b) Scheme for CuO/PEC nanocomposite synthesis, including its chemical structure scheme; (c) picture of prepared CuO/PEC solution; (d) CuO/PEC structural scheme and interactions illustration. (II) (a) CuO nanopowder and CuO/PEC NPs calculated release vs. time under 5.5 pH. (b) The process of Cu release from the CuO/PEC nanocomposite is depicted schematically. (III) Impact of CuO/PEC NPs on germination, (a) radicles (b) plumule (c) vigor index for radicles and plumule (d) radical length. (IV) Morphological advancements in tested seeds after 30 days treatment as compared to treatment with only water, PEC and CuO NPs. (Reproduced from ref. 111 with permission from Royal Society of Chemistry, 2021.)

high doses. Further it was found that although CuO allowed for high root development, and limited secondary root development, the CuO/PEC treatments significantly benefited both germination and root elongation, with a larger number of secondary roots as can be seen in Fig. 6(IV). It was evident that CuO/PEC 10 and CuO/PEC 50 showed the optimal root and shoot growth balance when compared to the ones treated with higher concentrations of MONPs in terms of how germination is dependent on their concentration (100 ppm). In fact, an excess of micronutrients is hazardous; in particular, high Cu concentrations inhibit root development, which leads to insufficient nutrient and water uptake. Besides, these harmful effects can be related to an increase in ROS production, which causes oxidative stress. A similar recent study revealed that

functionalized IONPs increased the yield, concentration and uptake of Zn in root, shoots and grain of wheat.¹¹⁷

Further, a recent study attempted to compare impact of chemically synthesized and green synthesized Fe₂O₃ nanorods (NRs) using *Moringa oleifera* along with bulk FeCl₃ on summer maize (*Zea mays*). It is reported that at concentrations of more than 25 mg L⁻¹, FeCl₃ salt and prepared Fe₂O₃ NRs induced growth retardation and compromised plant physiological and anti-oxidative activities due to toxicity by excess accumulation. Whereas, because of their low toxicity, Fe released from biologically generated NRs showed considerably good outcomes even at 50 mg L⁻¹. Furthermore, because of their ability to form complexes with metal ions, biologically produced NRs boosted plant anti-oxidative activity as well.¹¹⁸ Table 4 shows

Table 4 Biosynthesized MONPs based NFZs which are developed using various combinations of biological agents and metal precursors and investigated for their impacts against various target plants/seeds target pests when applied at different concentrations and morphologies

MONPs	Agents/precursors (concentrations) used in biosynthesis	Morphologies	NFZs concentrations/media	Target plants/seeds	Impacts	Ref.
Fe ₂ O ₃	Cornelian cherry (fruit extract): 20 g	Spherical	10–100 mg l ⁻¹	Barley seeds	• Increased shoot fresh mass	118 and 244–250
	Fe (NO ₃) ₃ ·9H ₂ O: 5 g	20 to 40 nm	Hoagland solution		• Much higher biomass accumulation	
	<i>Cassia occidentalis</i> L. flower extract: 20 ml	Irregular surfaces	NPs: 20 and 40 mg l ⁻¹	<i>Pusa basmati</i> rice seeds	• Higher rate of early & final germination	
	1 mMFeCl ₃ solution: 80 ml	20–50 nm	Medium-milli-Q water		• Increased Fe concentration in endosperm and embryonic tissues	
	Marine algae <i>Chaetomorpha antennin</i> : 0.1 g	10–18 nm	5–120 mg l ⁻¹	<i>Setaria italica</i> plants	• Concurrent rise in plant performance under drought stress	
	0.1 M FeCl ₂ ·4H ₂ O: 15 ml		Autoclaved distilled water		• Increased NPs concentration enhances soluble sugar content	
	Dried neem cake	200 nm to 2000 nm	2.08 mg l ⁻¹ NPs	<i>Vigna mungo</i>	• Post one week, plant reaches a maximum height of 17.8 cm	
FeSO ₄			Water		• Steadily increased alkaloids, glycosides, flavonoids, phenols, and steroids throughout the plant	
	<i>Paenibacillus polymyxa</i> DEBB B-358 containing auxin complex (IAA): 958 g	Spherical	50–100 mg l ⁻¹	Maize plantlets	• Increased germination rates, root growth, and fresh weight	
	20 mMFeCl ₃ ·6H ₂ O: 50 ml	26.65 nm	Distilled water			
	Beetroot (<i>Beta vulgaris</i>) leaf extract	—	4.0 mg l ⁻¹	Potato	• Increased early tuber induction and tuber growth	
	Iron salt		Distilled water			
	<i>Moringa oleifera</i> leaf extract	Crystalline nanorods	50 mg l ⁻¹	Summer maize (<i>Zea mays</i>)	• Increased plant anti-oxidative activity	
	FeCl ₃ ·6H ₂ O and FeCl ₂ ·4H ₂ O	16 ± 2 nm	Distilled water	Wheat plants	• Increased photosynthetic pigments and proline	
ZnO	Ginger (<i>Zingiber officinale</i>) and Cumin seeds (<i>Cuminum cyminum</i> L.) extracts	Crystalline ginger (0.6 mM) and cumin seeds (1.2 mM)	—		• Reduced lipid peroxidation	251–254
	—				• Excellent protection under drought stress	
	Black seeds (<i>Nigella sativa</i> L.)	Spherical	0–1000 mg l ⁻¹	Broccoli (<i>Brassica oleracea var italica</i>) seeds	• Improved seed germination by 37.5%, root length by 56.6%, and shoot length by 16.6%	
	ZnSO ₄ salt solution	24 nm	Distilled water		• Significantly increased Chl, phenolic acid, proline, and sugar levels	
	Brown algae (<i>Phaeophyta</i>) seed: 20 g	Spherical	25–150 mg l ⁻¹	Rice (<i>Oryza sativa</i> L.)	• Increased plant height, number of panicles, and panicle weight	
	ZnSO ₄ ·7H ₂ O: 14.73 g	31.4 nm	Double-distilled water			
	Basil extract: 100 ml	Spherical, poly faceted	3995.43 mg l ⁻¹	Basil (<i>Ocimum basilicum</i> L.)	• Increased fresh (32.57 g) and dry (10.85) weights	
MgO	ZnC ₄ H ₆ O ₄ salt: 100 ml	30 to 130 nm	Water			112
	Ginger (<i>Zingiber officinale</i>) and cumin seeds (<i>Cuminum cyminum</i> L.) extracts: 3 g	Spherical	0–500 mg l ⁻¹	Wheat	• Significantly increased length of both shoots and root	
	ZnC ₄ H ₆ O ₄	35–48 nm	DI water	<i>Triticum aestivum</i> L.		
MgO	<i>Aloe barbadensis</i> Mill (Aloe-Vera) leaf extract: 3 g	Spherically irregular	5 mg l ⁻¹	<i>Vigna radiata</i> (mung bean) and <i>Cajanus cajan</i> (red gram) seeds	• Increased seed germination rate of red gram by 100%	248
	Mg(NO ₃) ₂ : 10 g	23 nm			• Increased seed germination rate (mung bean seeds) by 90%	
MnO	<i>Paenibacillus polymyxa</i> DEBB B-358 containing auxin complex (IAA): 50 ml	Spherical	50–100 mg l ⁻¹	Maize plantlets	• Significantly increased germination rates, root growth and fresh weight	
	20 mM MnSO ₄ ·H ₂ O	22.32 nm	Distilled water			

recently developed biosynthesized MONPs-based NFZs for agricultural applications.

5.2. Nanopesticides (NPDs)

Plant diseases are caused by various pathogens such as bacteria, fungi, viruses, nematodes, insects, *etc.* In numerous recent studies, it was found that various MONPs have been shown to have high inhibitory activity against these pathogens.¹²⁰ Although many agrochemicals are currently on the market to work against phytopathogens, these conventional pesticides have poor colloidal dispersibility/stability, limited biological activity, and significantly less overall efficacy.¹²¹ In contrast, MONPs-based NPDs are more uniformly dispersed on crop leaves and have higher stability/solubility, making them easier for the active components of pesticides to reach the targeted organisms, thereby increasing their bio-

availability.¹²² Moreover, it has been reported higher efficiency of NPDs as they increased crop productivity by greater yields and reduced input expenses by lessening labour costs and waste.¹²³ Other advantages of NPDs have been depicted in Fig. 7(I). Furthermore, NPDs may be produced in two ways: directly as NPs (nanosized pesticides) or by coupling pesticides with nano-carriers in delivery systems. Pesticides are loaded into nano-carrier systems by encapsulation inside the nanoparticulate polymeric shell, absorption onto the nanoparticle surface, ligand binding to the nanoparticle core, or entrapment within the polymeric matrix⁹⁶ (Fig. 7(II)). In addition, mesoporous MONPs like SiO₂, ZnO, and TiO₂ can be used as physical adsorbents to adsorb active ingredients like pesticides through weaker interactions (hydrophobic and *van der Waals* forces), which can help to regulate the releasing rate of active ingredients and increase their half-life and stability.¹²⁴

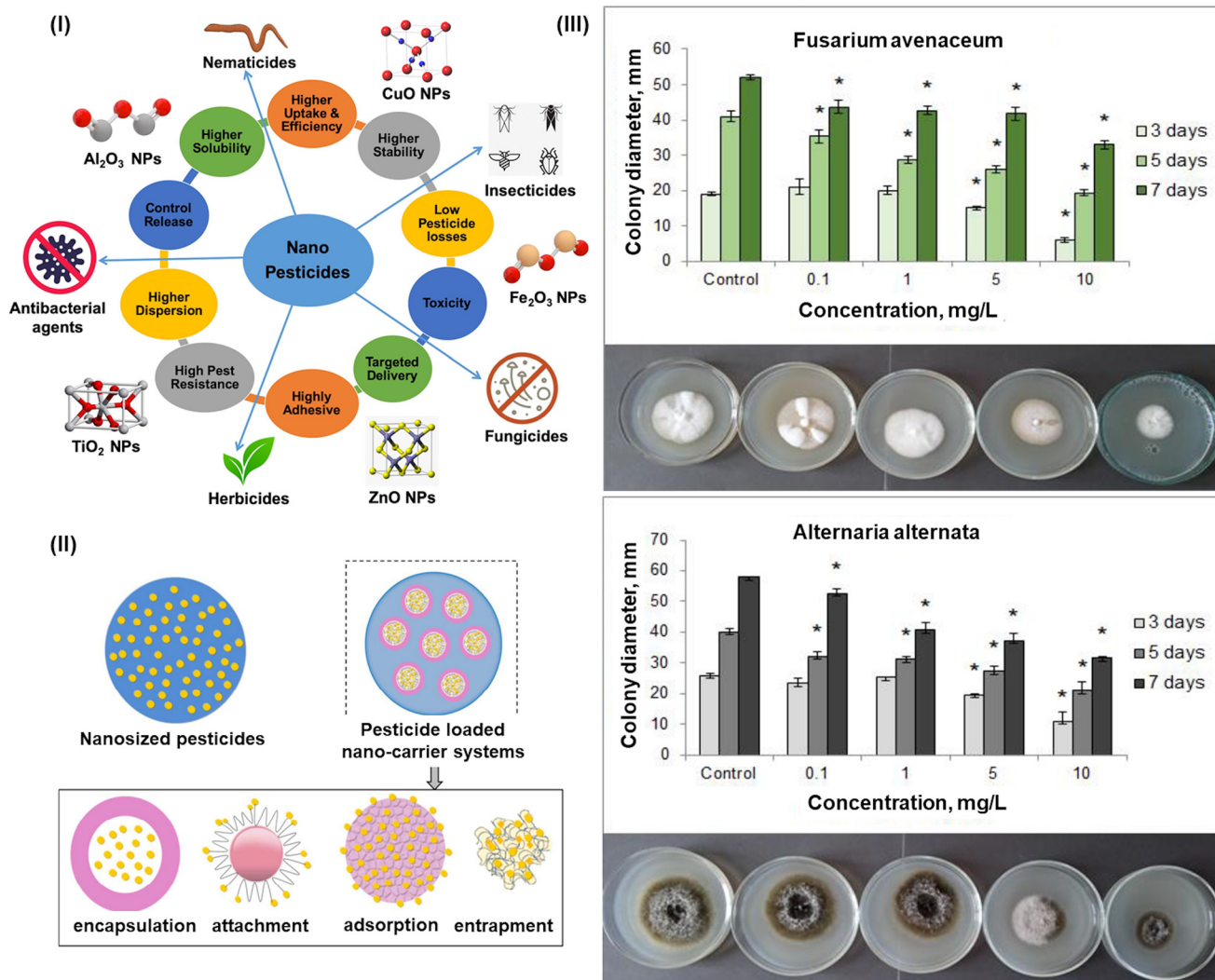


Fig. 7 (I) Various advantages of using MONPs as NPDs. (II) Schematic diagram of NPDs formulations (reproduced from ref. 96 with permission from American Chemical Society, 2022). (III) CuO NPs impact the growth of plant – pathogenic fungal colonies *Alternaria alternata* and *Fusarium avenaceum* on 3rd, 5th, and 7th day of growth in the presence of NPs (* is significant difference compared to control) (reproduced from ref. 119 with permission from MDPI, 2022).

Many researchers have investigated the use MONPs as NPDs in agriculture. For instance, in a latest investigation, it is reported that Fe₂O₃ NPs reduced the growth of *Tuta absoluta* (Lepidoptera) insect, which was inhibiting the development of tomato plants.¹²⁰ Another study confirmed the efficient larvicidal activities of IONPs when used against *E. coli* and *S. aureus* at 150 µg ml⁻¹ concentration.¹²⁵ Similar research demonstrated that ZnO NPs are also effective NPDs against *Bactericera cockerelli* when used on Tomato plants.¹²⁶ Further, it was evident by many reports that CuO NPs are one of the most promising candidates to be used as NPDs. For example, when used against cotton leafworms (*Spodoptera littoralis*), CuO NPs inhibited key enzymic activities of the pest through cell membrane disruption and eventually showed a high entomotoxic action at 232.75 mg L⁻¹ of exposure.¹²⁷ Similarly, when CuO NPs were used on tomato leaves against *Fusarium oxysporum* f. sp. *lycopersici*, 53.3% reduction in pest growth was observed.¹²⁸ Another recent study reported that CuO NPs effectively suppressed the development of colonies of phytopathogenic fungi *A. alternata* and *F. avenaceum* (up to 68.42% inhibition at 10 mg L⁻¹ CuO NPs) as depicted in Fig. 7(III).¹¹⁹ It was further revealed that CuO NPs reduced the biotic stress on *Birch clones* caused by these phytopathogens by gene regulations.

Furthermore, functionalized MONPs are being extensively investigated to deliver various pesticides efficiently and to cause lower toxicity. A recent study reported the application of unfunctionalized as well as amine-functionalized and polycaprolactone-coated Fe₃O₄ NPs to seed treatment in tomato.¹²⁵ It was revealed that the treatment has no side effects on plant germination and development. The results further demonstrated that all particles were taken up by the seeds, thus suggesting that the functionalized NPs can act as a versatile platform for delivering of active compounds, such as pesticides and growth factor agents. To minimize toxicity and further promote sustainability, biosynthesized MONPs are being investigated to be used as NPDs, for instance, in a recent study, bimetallic ZnO–TiO₂ NPs were synthesized biologically using fungal metabolites from *Metarhizium anisopliae* and tested for insecticidal activity.¹³¹ It was revealed that the bimetallic NPs showed the best larvicidal activity on *S. frugiperda* larvae at 100 µg ml⁻¹ after 48 h exposure. Pupicidal and anti-feedant activity was also observed in the tested NPs. The results strongly suggested that bimetallic ZnO–TiO₂ NPs synthesized using *M. anisopliae* can be used for protecting the crops against *S. frugiperda*. In a similar investigation, eco-friendly green synthesis of ZnO nanoparticles was done using the *Ulva lactuca* seaweed extract.¹³² The results demonstrated that ZnO NPs led to 100% mortality of *Aedes aegypti* fourth instar larvae at the concentration of 50 µg ml⁻¹ within 24 h. Another study assessed the acute toxicity of seven plants employed in ethno-veterinary medicine of southern India against third-instar larvae of *A. aegypti*.¹³³ Because *Lobelia leschenaultiana* leaf extracts had the highest efficiency against *Aedes aegypti* third-instar larvae, this plant species was employed for the production and stabilisation of ZnO NPs.

When evaluated at 10 mg L⁻¹, the *L. leschenaultiana*-encapsulated ZnO NPs demonstrated 100% death. Furthermore, a list of recently studied biosynthesized MONPs-based NPDs is given in Table 5.

5.3. Nano antimicrobial agents (AMBs)

The formation of more efficient AMBs is the need of the hour since present agrochemical treatments are costly and, in many cases, ineffective. The unique properties of MONPs, such as their ability to bind biomolecules, large surface area to volume ratio, high surface reactivity, low cytotoxicity, and ability to enhance gene expressions, allow them to be used as effective AMBs.¹³⁴ Recent advancements in biosynthesis have spurred scientists and researchers to harness its potential against numerous phytopathogens.¹³⁵ Biosynthesized Ag, Cu, Au, and Zn-based MONPs have been effective against many Gram-positive and Gram-negative bacteria such as *B. subtilis*, *E. coli*, and *Staphylococcus aureus*. These MONPs also inhibited the growth of pathogenic fungi such as *A. niger*, *F. oxysporum*, *A. fumigatus*, and other microbial pathogens.¹³⁶ Moreover, nanosize and variable surface chemistry of MONPs allow them to induce bacterial toxicity through several modes of action, such as lipid peroxidation, oxidative stress, cell membrane disruption, enzyme inhibition, proteolysis, and so on, as depicted in Fig. 8(I).^{137,138}

Many recent studies have reported effective antimicrobial activities of MONPs for instance, MgO NPs showed significant concentration-dependent antibacterial activity against phytopathogen *Ralstonia solanacearum* which was responsible for catastrophic bacterial wilt and caused a worldwide reduction in tobacco production.¹³⁹ Further, the use of biological components such as plants and microorganisms for MONPs production has been investigated in order to discover plants with high inhibitor concentrations and diverse phytochemical compositions that may be employed in effective antibacterial applications. In a recent study, enhanced antibacterial activity against *S. typhimurium* was shown by ZnO NPs, which were biosynthesized using saffron leaf extract.¹⁴⁰ It was revealed that to reduce the impact of hazardous chemicals used in NPs' synthesis, saffron leaf extract can be utilised as a reducing agent or a stabiliser. Moreover, ZnO NPs' antibacterial activity is ascribed to their amphoteric nature, which allows Zn²⁺ ions to penetrate and break down the bacterial cell membrane. Similar investigations were reported for antimicrobial activities of biosynthesized ZnO NPs using *Aegle marmelos* leaves extract and *Cassia alata* extracts and revealed dose-dependent high percentage inhibition against *A. niger* and *Escherichia coli*, respectively.^{141,142} In another investigation, ZnO NPs synthesized using *Eucalyptus lanceolatus* (leaf litter) showed effective antimicrobial activity against pathogens causing diseases in maize plants.¹⁴³ Analogously ZnO and TiO₂ NPs developed with lemon fruit extract demonstrated antibacterial activity against *Dickeya dadantii*, the causative bacterium of sweet potato stem and root rot disease.¹⁴⁴ Further, fourier transform infrared spectroscopy demonstrated that the functional groups on the surface of the ZnO and TiO₂ NPs, which

Table 5 Biosynthesized MONPs based NPDs which are developed using various combinations of biological agents and metal precursors and investigated for their impacts against various target pests when applied at different concentrations and morphologies

MONPs	Agents/precursors (concentrations) used in biosynthesis	Morphologies	NPDs concentrations/media	Target pests	Impacts	Ref.
TiO ₂	<i>Propolis</i> extract: 10 ml of extract (500 mg ml ⁻¹) 2 mM Ti(OH) ₂ soln.	57.5 nm Cubic to rectangular shape	32–64 mg l ⁻¹ Distilled water	<i>P. vulgari</i> , <i>S. epidermidis</i>	• Showed efficient pesticidal effects at high concentrations • High chemical stability and low toxicity in humans	255
CuO	<i>Aspergillus Niger</i> Fungi Strain: 15 g with 100 ml distilled water 2 mM CuSO ₄ ·5H ₂ O	14–47.4 nm Crystallographic-spherical	100–200 mg l ⁻¹	<i>Sitophilus Granarius</i> , <i>Rhyzopertha Dominca</i>	• High efficacy against pests	256
Fe ₂ O ₃	<i>T. foenum-graecum</i> Leaf extract: 10 ml 0.1 M FeCl ₃ : 1.5 ml <i>Celosia argentea</i> Leaf Extract: 10 ml 0.1 M ferric nitrate soln	3.8–11.95 nm Irregular shape 5–10 nm Spherical 2–8 nm	5 to 100 mg l ⁻¹ Distilled water 150 mg l ⁻¹ Distilled water 149.82 mg l ⁻¹	<i>Tuta absoluta</i> (Lepidoptera) <i>Anopheles subpictus</i> (Larvae) <i>S. oryzae</i> and <i>S. cerealella</i>	• Low toxicity • Enhanced crop growth • Showed strong insecticidal activity at very low concentrations • Efficient insecticidal nature	257 and 258
ZnO	Pomegranate Peels 1 M Zinc Nitrate soln <i>Nyctanthes Arbor-Tristis</i> Flower Extract soln: 1 ml 0.01 M Zinc Acetate soln.: 50 ml <i>Eclipta alba</i> extract 10 ml. ZnSO ₄ (10 mM): 80 ml	Spherical 2–8 nm Spherical Crystalline 12–63 nm Hexagonal Crystalline Polydisperse nature 32 nm Hexagonal Mondispersenature Wurtzite structure	Distilled water Distilled water Distilled water 256 mg l ⁻¹ Sterile deionized water 49.94 mg l ⁻¹ Sterile distilled water	<i>S. oryzae</i> and <i>S. cerealella</i> <i>Carbendazim</i> and <i>thiram</i> <i>S. graminicola</i>	• Showed progressive increase in mortality of pest adults by increasing exposure intervals and concentration • Better inhibition of test fungi than chemically synthesized NPs • Reduced downy mildew by 35% • Promoted growth and induced systemic resistance	30, 259 and 260

were responsible for NPs' stabilisation, were derived from lemon fruit extract. In another study, MgO and MnO₂ NPs synthesized using chamomile flower extract displayed effective inhibition in the growth of *Acidovorax oryzae* strains (a bacterium that causes rice bacterial brown stripe disease).¹³⁶ From all these latest studies, it is evident that MONPs possess great potential to be used in plant pathogenic control.¹⁴⁵ Moreover, a comparative study of antimicrobial activities of chemically and biosynthesized FeO NPs suggested that both the FeO NPs exhibited potential antibacterial activity but biosynthesized FeO NPs exhibited comparatively higher bactericidal activity. It was revealed that biosynthesized FeO NPs could render the cells non-viable at concentration of 100, 100 and 200 µg mL⁻¹ for *Escherichia coli*, *Pseudomonas aeruginosa* and *Klebsiella pneumonia*, respectively. Whereas, chemically synthesized FeO NPs rendered non-viability at the concentrations of 200, 400 and 400 µg mL⁻¹ respectively, clearly indicating that biosynthesized MONPs are more efficient.⁴⁹ In a similar comparative study of chemically synthesized ZnO NPs and biosynthesized ZnO NPs using *Juglans regia* L. leaf extract again demonstrated that green ZnO NPs showed better antimicrobial activities.¹⁴⁶ Additionally, in another study, dependence of the antimicrobial action of IONPs on the size and type of iron oxides was estimated by assessing various relevant literature.¹³⁰ Based on the analyzed literature data it was observed from Fig. 8(IIa) that same sized green IONPs showed better antimicrobial

activities compared to their chemically synthesized counterparts. Further meta-analysis revealed that Fe₂O₃ NPs show more pronounced bacteriostatic actions compared to Fe₃O₄ NPs (Fig. 8(IIb)) because Fe₂O₃ NPs can bind directly with the bacterial cell wall and further penetrate and concentrate in the cytoplasm, causing vacuole formation and cell wall breakdown. Lastly, Fig. 8(IIc) demonstrates that IONPs generated by the biosynthesis approaches had three times more pronounced bacteriostatic activity than IONPs generated by the coprecipitation method. Furthermore, a list of recently developed biosynthesized MONPs and their size; shape; and target microbial pathogen are mentioned in Table 6.

5.4. Nanobiosensors (NBSs)

NBSs can detect nutrients, toxins, and pathogens in the soil to safeguard soil health, environment, and fertility. They can also determine a plant's nutritional, moisture, and physiological state.¹⁴⁷ Due to their specificity, portability, rapid response time, and low cost, NBSs are projected to play a critical role in clinical and non-clinical applications.¹⁴⁸ Following are the three essential stages of NBSs development (illustrated in Fig. 9(A)) which are further demonstrated with an example in which DNA of *S. enteritidis* pathogen was detected by a specific DNA bioreceptor of Au/Fe₂O₃-DNA NBS as recorded by an electrochemical transducer in a differential pulse voltammogram without the use of an amplifier (Fig. 9(B)).¹⁴⁹

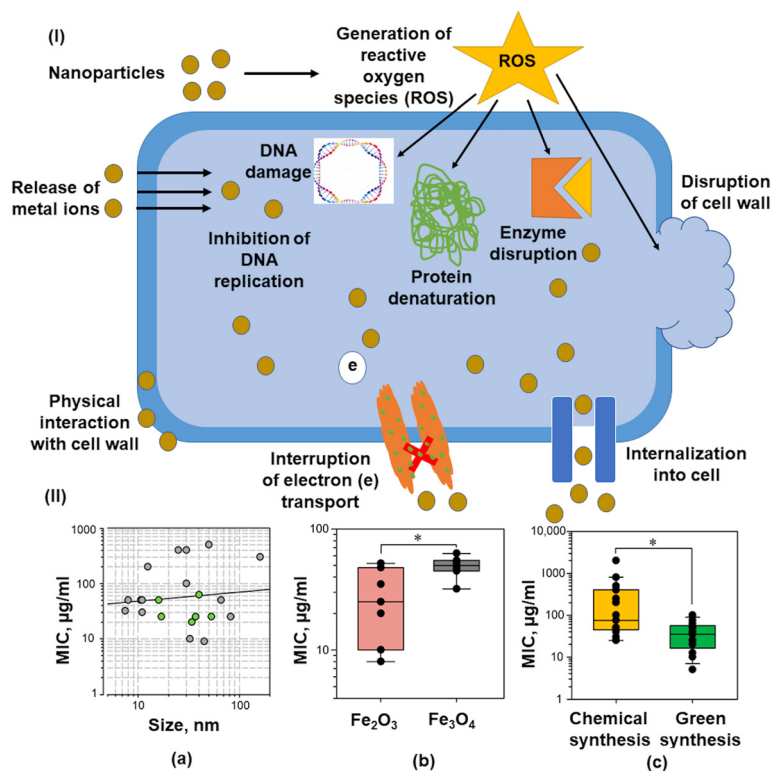


Fig. 8 (I) Various mechanisms responsible for the antimicrobial activities of MONPs. MONPs can generate ROS, release ions, disrupt cell walls, and further internalize into cells of microbial pathogens, causing DNA damage, inhibition of DNA replication, enzyme disruption, and protein denaturation (damaging the molecular structure of protein).¹²⁹ (II) (a) Evaluation of the IONPs' MIC's reliance on IONPs' size versus *E. coli*. (b) Evaluation of the IONPs' MIC against *E. coli*'s reliance on iron oxide type. (c) The dependency of the IONPs' MIC against *E. coli* on the synthesis technique. In (a), the grey hue represents the IONPs acquired without "green synthesis", whereas the green colour represents the IONPs produced with "green synthesis" (* $p < 0.05$) (reproduced from ref. 130 with permission from MDPI, 2021).

I. Sample preparation: It could be a homogenous or complex suspension of gas, liquid or solid-state. Sample preparation is difficult for the agroecosystem because of various impurities and interferences. The sample consists of specific molecules, functional groups of molecules or organisms, for the sensors to target. Targeted molecules/organisms are called the analyte and are generally molecules (dyes/colors, toxicants, pesticides, hormones, antibiotics, vitamins, etc.), biomolecules (enzymes, DNA/RNA, allergens, etc.), ions (metals, halogens, surfactants, etc.), gas/vapor (oxygen, carbon dioxide, volatile compounds, water vapors, etc.) and organisms (bacteria, fungi, viruses).¹⁵⁰

II. Recognition: The analytes in the sample are identified by specific molecules/elements. These recognition molecules include antibodies, aptamers, chemical legends, enzymes, and others that have high affinity, specificity, and selectivity to their analytes to measure them to acceptable levels.¹⁵¹

III. Signal transduction and amplification: Different signal transduction methods have classified these small devices as optical, electrochemical, piezoelectric, pyroelectric, electronic, and gravimetric biosensors.¹⁵² They translate recognition events such as changes in current, colour, plasmonic characteristics, fluorescence emission intensity, and so on into computable signals, which are then processed to provide the data.

These signals are usually weak, and hence an amplifier is attached for proper detection.¹⁵²

Further, the recent advances in nanotechnology have opened various new ways for the hybridization of NPs (such as carbon nanotubes, quantum dots, metal, and MONPs) with different biosensing devices,¹⁵³ offering a great deal of conjoining and multipurpose approaches for enhanced sensitivity for detection and thereby improving the capability of monitoring even a single molecule. Moreover, NBSs are reproducible, affordable, and robust, thereby making them a further attractive alternative.¹⁵⁴ In this section, we will discuss different MONPs-based NBSs.

As mentioned before, NBSs are used to detect pesticides and heavy metals from the soil due to their extraordinary sensitivity characteristics. For example, Ta₂O₅ NPs was used to detect fenitrothion pesticide due to the change in refractive index, using a surface plasmonic resonance-based fiber-optic transducer.¹⁵⁵ In another study, a hybrid nanocomposite of CuO-TiO₂ detected methyl parathion (a widely used organophosphate insecticide) with the help of an electrochemical sensor by assessing differential pulse voltammetry measurements.¹⁵⁶ Further, in another investigation, Malathion pesticide was detected by Chitosan-Fe₂O₃ NPs and CuO NPs with the help of an electrochemical transducer using differential

Table 6 Biosynthesized MONPs as AMBs which are developed using various combinations of biological agents and metal precursors and investigated for their impacts against various target pathogens when applied at different concentrations and morphologies

MONPs	Agents/precursors (concentrations) used in biosynthesis	Morphologies	AMBs concentrations/media	Target pathogens	Impacts	Ref.
CuO	<i>Penicillium chrysogenum</i> : 5 mL	9.70 nm	$250 \times 10^3 \text{ mg l}^{-1}$	<i>Fusarium oxysporum</i> , <i>alternaria solani</i> , <i>aspergillus niger</i> , <i>ralstonia solanacearum</i> , <i>erwinia amylovara</i>	• Maximum antifungal activity against pathogens • Strong bactericide and fungicide	210 and 261–264
	CuSO ₄ ·5H ₂ O solutions (1.0, 5.0 and 10.0 mM): 5 mL	Spherical	Distilled water			
	<i>C. papaya</i> L. leaf extract: 10 mL	40 to 80 nm	50, 125, and 250 ($\times 10^3 \text{ mg l}^{-1}$)	<i>Ralstonia solanacearum</i>	• Reduced bacterial cell viability • Inhibited bacterial biofilm formation • Reduced bacterial swarming motility • Strong antibacterial activity • Distorted and destroyed the cell membranes; caused death of bacterial cells • Increased zones of inhibition with NPs concentrations • Effective antibacterial activities • Increased inhibitions zones	
	5 mM CuSO ₄ : 90 mL	Irregular spherical	0.35% agar medium			
	<i>O. tenuiflorum</i> leaf extract: 10 ml	Spherical as well as thick-needle morphologies	5–15 ($\times 10^3 \text{ mg l}^{-1}$)	<i>Escherichia coli</i> , <i>Streptococcus mutans</i> , <i>Proteus vulgaris</i> , and <i>Staphylococcus aureus</i>		
	Cu(CH ₃ COO) ₂ solution: 0.2 M		Agar plates			
	Leaf extracts of <i>Eupatorium odoratum</i> and <i>Acanthospermum hispidum</i> : 10 mL	50 nm	100 mg l ^{−1}	<i>Staphylococcus aureus</i> , <i>Bacillus cereus</i> , <i>Escherichia coli</i>		
	Aqueous solution of CuSO ₄ : 5 mL	Spherical	Distilled water		• Effective antibacterial activities • More effective against Gram-negative bacteria • Effective for bacterial blight treatment of rice • Effectively increased inhibition with NPs concentration • Inhibited <i>in vitro</i> bacterial growth and controlled the bacterial wilt of tomato • Good antibacterial activity • Effectively increased inhibition with NPs concentration • Efficient bactericidal activity • Plant extracts also possess antibacterial activity due to the presence of phytochemical components • Potentially useful as antifungal and antibacterial compound	
	<i>Annona squamosa</i> aqueous seed extract	11 nm	500 and 1000 mg l ^{−1}	<i>Escherichia coli</i> , <i>Microbacterium testaceum</i> , <i>Xanthomonas oryzae</i>		
	CuSO ₄ 1 : 1 ratio	Spherical				
Fe ₂ O ₃	Aqueous <i>Skimma laureola</i> leaf extract: 0.1 M	Crystalline	1, 3, 6 ($\times 10^3 \text{ mg l}^{-1}$)	<i>R. solanacearum</i>		25, 208 and 265–267
	FeCl ₃ ·6H ₂ O: 0.1 M	34 ± 0.37 nm				
		Crystalline Cubic	Nutrient Agar media			
	<i>Eichhornia crassipes</i> leaf extract: 100 mL	Rod shaped	25 and 100 mg l ^{−1}	<i>S. aureus</i> , <i>P. fluorescens</i> , <i>E. coli</i>		
	0.1 M FeSO ₄ : 100 mL		Deionised water			
	<i>Ruellia tuberosa</i> leaf extract: 50 mL	52.78 nm	25, 50 and 75 mg l ^{−1}	<i>K. pneumonia</i> , <i>E. coli</i>		
	1 M FeSO ₄ : 50 mL	Hexagonal nanorods	Milli-Q water			
	<i>L. nobilis</i> leaf extract: 30 mL	8.03 ± 8.99 nm	—	BACTERIA: <i>Listeria monocytogenes</i>		
	0.1 M FeCl ₃ ·6H ₂ O: 30 mL	Crystalline		FUNGUS: <i>Aspergillus favus</i> and <i>Penicillium spinulosum</i>		
		Almost spherical like and partly as a hexagonal shape				

Table 6 (Contd.)

MONPs	Agents/precursors (concentrations) used in biosynthesis	Morphologies	AMBs concentrations/media	Target pathogens	Impacts	Ref.
	5 ml 0.004 M tannic acid	10–30 nm	0.1, 0.2, 0.3, 0.4, 0.5 ($\times 10^3$ mg l ⁻¹)	<i>Trichothecium roseum</i> , <i>Cladosporium herbarum</i> , <i>Penicillium chrysogenum</i> , <i>Alternaria alternata</i> and <i>Aspergillus niger</i>	• Significant antimycotic activity	
MnO ₂	0.5 g Fe dissolved in 50 ml double distilled water	Crystalline	Deionised water		• Recorded highest activity index against <i>P. chrysogenum</i>	
	<i>Chamomile</i> flower extract 1 M MnO ₂ 1 : 1 ratio	Hexagonal 36.6 to 112.0 nm Irregularly spherical shaped	16.0 mg l ⁻¹ Distilled water	<i>A. Oryzae</i> strain rs-2	• Strong antibacterial effects • Reduced the growth by 71.3% • Significantly reduced biofilm formation and swimming motility • Completely suppressed biofilm formation by 74.4%	268 and 269
	<i>Paenibacillus polymyxa</i> strain Sx3 bacteria extract: 100 mL MnO ₂ : 100 mL	18.8 nm Spherical	16.0 mg l ⁻¹ Distilled water	<i>Xanthomonas oryzae pv. Oryzae</i>	• Excellent antibacterial efficacy • Good against bacterial leaf blight disease • Increased rice biomass. • Completely suppressed growth at 15.36 μ g ml ⁻¹	270 and 271
	Bacterial cell filtrate of the endobacterium <i>B. rinojensis</i> 0.1 M [Mg (NO ₃) ₂] solution	26.70 nm Roughly spherical granular structure	15.36 mg l ⁻¹ , 1.92 mg l ⁻¹ Distilled water	<i>Fusarium oxysporum</i>	• Inhibited formation of biofilm completely at 1.92 μ g ml ⁻¹ • Significant damaged fungal membrane integrity • Significantly inhibited bacterial growth, biofilm formation, and motility • Destroyed cell integrity, resulting in leakage of intracellular content	
ZnO	<i>Rosmarinus officinalis</i> L. (Rosemary) extract: 100 Ml	8.8 nm	4, 8, and 16 mg l ⁻¹	<i>Xanthomonas oryzae pv. Oryzae</i>	• Reduced biofilm formation and swimming • Damaged cell membrane • Increased ROS production • Increased apoptosis of bacterial cells	272 and 273
	Aqueous solution of 1.0 mM MgO solution: 100 mL	Hexagonal crystal structure	Distilled water		• Significantly inhibited bacterial growth, biofilm formation, and motility • Destroyed cell integrity, resulting in leakage of intracellular content	
	Aqueous green tomato extract: 100 Ml	31.3 to 88.9 nm	4.0, 8.0, and 16 mg l ⁻¹	<i>Xanthomonas oryzae pv. Oryzae</i>	• Reduced biofilm formation and swimming • Damaged cell membrane • Increased ROS production • Increased apoptosis of bacterial cells	
	1 M ZnO: 100 mL	Spherical and hexagonal shapes Crystalline	Distilled water		• Significant antibacterial activity at 50 μ g ml ⁻¹ with a 2.83 cm and 2.18 cm zone of inhibition • Reduced cell numbers by 71.2% and 68.1% • Caused morphological damage to the pathogen	
	<i>Native Bacillus cereus</i> RNT6 strain extract: 100 mL	21 to 35 nm	50 mg l ⁻¹	<i>Burkholderia glumae</i> and <i>B. Gladioli</i>		
	M ZnSO ₄ ·7H ₂ O: 100 mL	Spherical Crystalline				

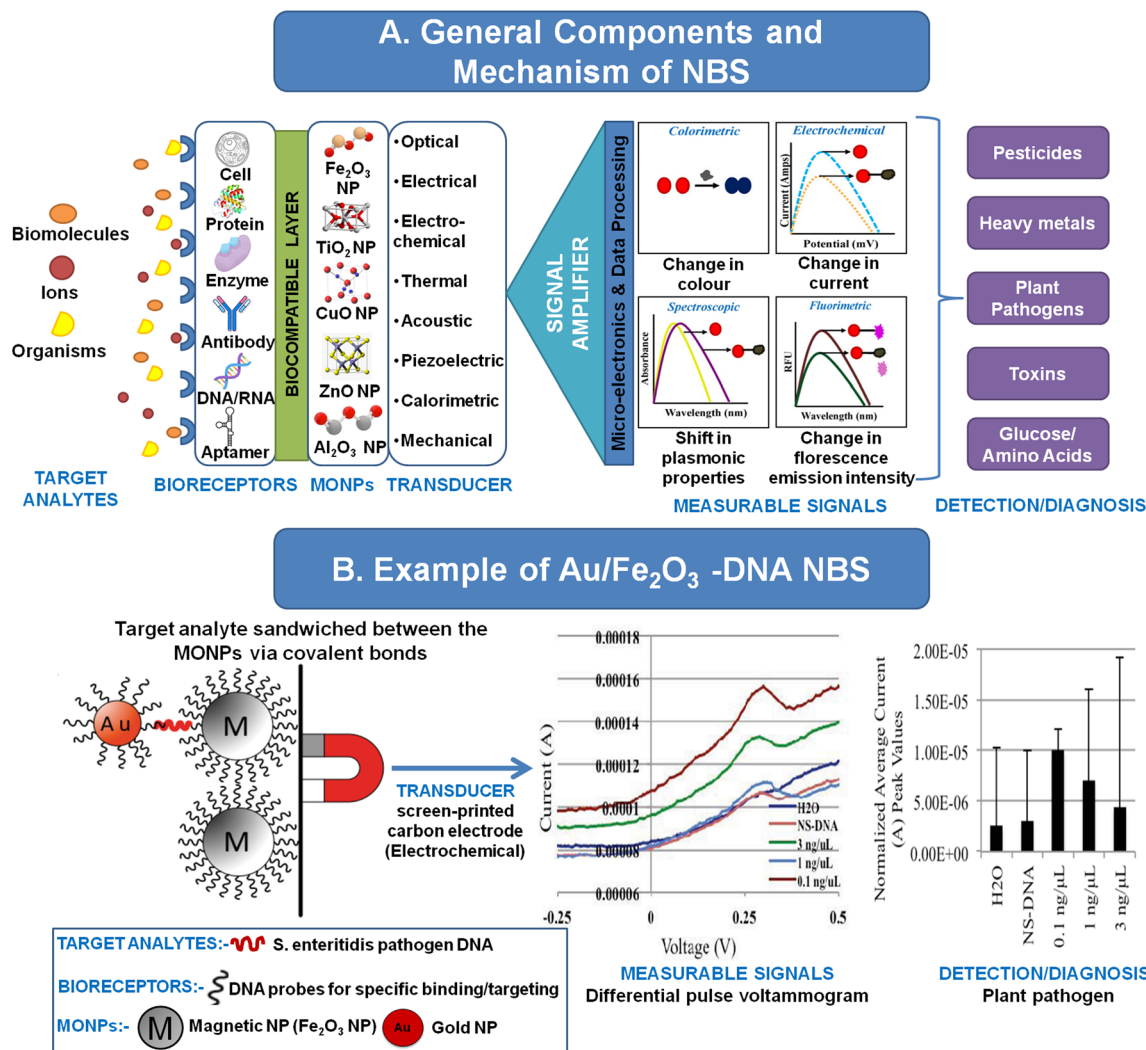


Fig. 9 (A) The mechanism and components of NBSs are illustrated with the help of an example. Firstly, target analytes (specific biomolecules, ions, or organisms for the sensors to target) are recognized by further bioreceptors (cell, protein, enzyme, DNA, antibody, etc.). Certain recognition events such as changes in the current colour, plasmonic characteristics, and fluorescence emission intensity are translated into computable signals with transducers, and various components like pesticides, heavy metals, pathogens, etc., can be detected. (B) Example: DNA of *S. enteritidis* pathogen was detected by a specific DNA bioreceptor of Au/Fe₂O₃-DNA NBS as recorded by an electrochemical transducer in a differential pulse voltammogram without needing an amplifier (reproduced from ref. 149 with permission MDPI, 2012).

pulse voltammetry assessments.^{157,158} Another study reported that superparamagnetic Fe₂O₃ NPs detected Hg *via* fluorescence quenching with increasing Hg²⁺ concentration. The detected analyte was removed by an external bar magnet, hence, there was no residual pollution.¹⁵⁹ Although other methods such as liquid–solid extraction, gas chromatography, high-performance liquid chromatography, mass spectrometry, and so on can detect these pesticides and heavy metals with high sensitivity and reliability, they are time-consuming, require tedious pre-treatment procedures, are highly expensive, and require trained personnel; thus, NBSs are being developed for more convenient, direct, quick, and sensitive detection for efficient environmental monitoring.¹⁶⁰ Similarly, NBSs are widely being formulated to easily and effectively detect phytopathogens. In a recent study, Fe₃O₄/SiO₂ NPs detected tomato ringspot virus, bean pod

mottle virus, and *Arabidopsis mosaic virus*, by recognizing an excellent linear relationship between enhanced fluorescence and virus concentration.¹⁶¹ Likewise, TiO₂/SnO₂ NPs on screen-printed electrodes used an electrochemical sensor to detect *Phytophthora cactorum* by identifying symbolic volatile compound *p*-ethyl guaiacol (produced during infection).¹⁶²

Further, NBSs have been established for detecting some crucial molecules like amino acids, glucose, and H₂O₂, owing to their chemical steadiness, bio-compatibility, and easy surface alteration. For instance, in the latest study, SnO₂-based NPs with particle size 4 nm was used to detect L-systeine amino acids. These NPs were synthesized by precipitation, sonication & microwave irradiation.¹⁶³ In another study, Fe₂O₃-based NPs, also having a particle size of 4 nm, displayed effective results for sensing H₂O₂.¹⁶⁴ Similarly, TiO₂-based NPs

having a particle size of 3 nm was used for glucose detection.¹⁶⁵ In another research, MnO₂-based biosensors were fabricated using MnO₂ particles *in situ* deposited on the glucose oxidase surface. These electrodes show high electrocatalytic activity for H₂O₂ detection and had a sensitivity and detection limit of around 38.2 $\mu\text{A cm}^{-2} \text{mm}^{-1}$ and 0.8 μm , respectively.¹⁶⁶ In another recent investigation, CuO-ZnO nanocomposites and nanofibres were calcinated and used as biosensors due to low detection limit, high sensitivity, and stability for the non-enzymatic detection of glucose.¹⁶⁴ Similar research demonstrated that carbon-doped TiO₂ nanotubes arrays detected 5-hydroxytryptamine and ascorbic acid having a concentration between 5 to 150 μm and up to 4.1×10^{-8} M limit.¹⁶⁷

Furthermore, magnetic NPs (MNPs) can be used as bio-recognition probes for the selective capture of any desired analyte from complicated real-world sample matrices (e.g., bio-fluids such as whole blood, serum, urine and saliva, tissues, food, and environmental samples).¹⁶⁸ MNPs-based biosensing technique is extremely appealing due to the quick separation of captured analytes by an externally applied magnetic field whereas, other inorganic particles need complex centrifugation and separation procedures. Moreover, MNPs-biosensors have now enabled ultra-low detection with ultra-high sensitivity, which was previously only possible with radioactive assays and other tedious methods.¹⁶⁹ Similar technique namely magnetic particle spectroscopy (MPS) has evolved in recent years into a hypersensitive and adaptable sensing tool for quantitative bio-assays. It detects target analytes such as viruses, toxins, nucleic acids, and so on using the dynamic magnetic responses of MNPs. For instance, a recent report has demonstrated an efficient biosensing application of MPS to monitor the uptake and distribution of different sized (10 and 20 nm) chelated IONPs in garden cress (*Lepidium sativum*) plants and shown how to overcome the challenge of distinguishing intact IONPs from leached Fe ions during their investigation.¹⁷⁰ Additionally, many such MNPs-based environmental biosensor devices have been developed recently for detecting pathogens, pesticides, and hazardous soil toxins. For instance, co-immobilization of carboxylic acid group-modified MNPs and acetylcholinesterase enzyme onto an electropolymerized surface of 4,7-di(furan-2-yl) benzo[c][1,2,5]-thiadiazole enabled sensitive micromolar detection of paraoxon and trichlorfon which are significant components of hazardous organophosphorus pesticides.¹⁷¹ In a similar study, acetylcholinesterase-MNPs based electrochemical biosensor was developed which could detect organophosphate pesticides with high sensitivity.¹⁷² In another investigation, *Escherichia coli* was detected quickly and accurately using an impedimetric electrochemical aptasensor.¹⁷³

Moreover, toxic reagents such as sodium borohydride, ascorbic acid, and hydrazine hydrate are used in the chemical production of NPs. However, the use of toxic chemicals in the aforementioned processes limits NPs' biocompatibility and applicability. As a result, green synthesis of NPs has been sparked, and their prospective application as NBSs has gained substantial interest for sustainable agricultural and environmental monitoring.

6. Current challenges and future prospects

The most cost-effective and environmentally friendly technique to produce finely scaled MONPs has been found to be the biosynthesis method for their usage in the agricultural sector as growth stimulants and management tools. However, there are several concerns with this approach, such as the formation of polydispersed NPs *via* plant-mediated synthesis, their less colloidal stability due to agglomeration, the excessive time required for fungal-mediated synthesis, and the lack of data on algal-mediated synthesis.¹⁰⁶ Besides, reproducibility of this methodology is another challenge. Further, due to this technique's passive approach and the corresponding reactants' incomplete utilization, the reaction in this process produces undesirable products which might cause harm to the plants when used for protection. Hence, many researchers have now been working on this issue to remove these toxic biproducts during the synthesis steps. Additionally, the idea of parameters governing the size, shape, magnetic characteristics, and yield of MONPs is being researched. It is required to produce size and shape-controlled synthesized NPs. Also, identifying the major biomolecules involved in metal ion reduction can provide the groundwork for developing a more optimal and efficient biological synthesis strategy.¹⁰⁶ Green synthesis/biosynthesis is developing as a better alternative to the conventional methods used earlier but further steps can be taken to enhance its economical feasibility as well. For instance, using organic waste from food and agriculture as a biological supply of reducing agents for MONPs synthesis can significantly reduce manufacturing costs.^{106,174} Moreover, bionanominig is a new strategy that involves using mine water as a source of metal precursors for the synthesis of metallic nanoparticles, which has the potential to minimize the usage of concentrated chemicals for the preparation of metal salt solutions, thereby boosting economic feasibility.^{106,175} Moreover, MONPs play a significant part in the agriculture industry as it has an amalgamation of the benefits of metal and exceptional properties of NPs. Despite the existence of numerous studies on the potentials of nanotechnology in crop fertilisation, protection, and management, commoditization of these nanoproducts is highly restricted which may be due to different challenges and shortcomings surrounding their use.

Furthermore, it is important to discuss the fate of MONPs if left in the environment without monitoring. A few recent studies have evaluated the toxicity of NPs to the environment and human health. Numerous mechanisms, including membrane rupture, protein oxidation, genotoxicity, interruption of energy transmission, generation of ROS, and release of hazardous substances are associated with the toxicity of NPs. Due to their high reactivity (associated with smaller size and large surface area), surface charge, chemical composition, and reactive hydrophobic/lipophilic groups (that may interact with plant proteins/membranes), NPs may contribute to enhanced toxic effects on plants and inhibit the activity of plant enzymes

through bioaccumulation. Studies reporting the impact of NPs on complex systems such as soil microflora open the door to risk evaluations connected with their interaction with the environment. Soil fertility is directly affected by the microorganisms present in the soil since they play an important role in the biogeochemical cycle, decomposition of soil organic matter, and plant development. It has been observed that the release and accumulation of engineered NPs in soil, as well as their interaction with soil microflora, have a negative impact on microbial diversity.^{176,177} A recent study indicated the harmful influence of Al₂O₃ NPs on soil biocenosis, which resulted in soil deterioration and reduced fertility.¹⁷⁸ Moreover, a number of studies suggest that the presence of Ag and its salts in soil significantly impairs the nitrogen cycle.^{179–182} Many other studies reveal that NPs harm beneficial microbial cells through a variety of exterior and interior toxicity mechanisms. Additionally, NPs have an inhibiting effect on unicellular, mycorrhizal, and cellulose/lignin-degrading fungi. For instance, TiO₂ and positive polystyrene NPs reduced rhizosphere bacteria, as reported in recent studies.^{183,184} Although there are several studies suggesting that prolonged exposure of soil microflora to MONPs such as ZnO and CuO causes no substantial changes in enzymatic activity or population number of the examined group of microbes¹⁸⁵ but it is extremely necessary to perform more such assessments to reveal further about the toxicity and fate of NPs towards the environment.

Besides, many studies have been reporting that MONPs cause numerous side effects on human health. A recent study has revealed that TiO₂, SiO₂, and ZnO NPs have the ability to impair intestinal function by changing the abundance of gut bacterial populations.¹⁸⁶ Another study suggests that MONPs have the potential to provoke a robust immunological response, which can include immunosuppression, inflammation and autoimmunity, as well as sensitivity and hypersensitive responses. Further, NPs can induce immune-mediated human health concerns resulting from cellular mechanisms of damage such as oxidative stress, interactions with cellular membranes and proteins, and alteration of signaling cascades.¹⁸⁷ More investigations confirmed the negative effects of ZnO and CuO NPs on human health. For instance, CuO NPs were reported to cause mitochondrial dysfunction in human umbilical vein endothelial cells, lysosomal damage in human blood lymphocytes and increased ROS levels.^{188–190} Similarly, many reports suggest cytotoxicity of ZnO NPs for breast cancer causing membrane blebbing and reduced cell viability due to oxidative stress caused by ROS.^{191–194} These results imply a need for more such assessments and identification of cautionary measures for using nanotechnology in agriculture. Currently, most studies emphasise that NPs only exhibit toxicity when used in large doses and numerous studies are making progress in figuring out the ideal concentration of MONPs to be applied to crops for sustainable agriculture. However, it is essential to note that the toxicity of NPs is not just attributed to their concentration but is also determined by their types/composition, size distribution, aggregation, surface

charge, reactivity and surface modification/functionalization. Moreover, the entire toxicological consequences of NPs on various biological systems such as fungi, insects, and mammals are also not entirely known.¹⁹⁵ Hence, with the increasing demand for nanoproducts, it is an absolute need of the hour to develop proper regulations and safety measures for applying nanotechnology, especially in agriculture.

Significant institutions like the United States Department of Agriculture (USDA) have also acknowledged the use of nanotechnology in agriculture and food systems in their latest article and further emphasised the understanding of potential risks of NPs which will help make policy decisions related to their manufacturing and usage.¹⁹⁶ It was further mentioned that the University of Florida, the University of Kentucky, and the University of Wisconsin are researching the human and environmental health consequences of nanotechnology in food, agricultural, and biological systems. Moreover, University of Kentucky researchers were among the first to investigate and demonstrate the intergenerational impacts of Ag NPs exposure. Other challenges linked with agricultural nanotechnology include limited investments in R&D infrastructure, high production costs, low agricultural returns, and inefficiencies in technology transfer and implementation in the agricultural sector, which hinder their advancements.¹⁹⁷

Further, we must keep in mind that in 2050, the global population is predicted to reach 9.6 billion or more. To serve the world's ever-increasing population, undue pressure will be imposed on land, which cannot be expanded. Conventional agrochemicals may not be useful since they are expensive due to their high energy demands and are environmentally hazardous. Besides, after witnessing the indubitable potential of MONPs to cater to all these needs through this review, it is only fair to say that biosynthesized MONPs-based agricultural nanoproducts in sustainable agriculture will inevitably rise based on their positive results through enduring tons of research in the upcoming future. Moreover, different mechanisms associated with NPs-plant interaction are necessary to be further proven effectively since many of its aspects, like ROS generation and translocation pathways, are only hypothesized. However, further lab-scale studies discovering concrete evidence of these theories are essential to understanding the long-term impact of MONPs on the environment and are bound to increase in the future. To understand the fertilisation and plant-protection processes, studies comprising three-way comparisons of nanoformulations, conventional treatments, and active ingredients are highly suggested and measuring the effectiveness of new products *versus* current formulations. Furthermore, large-scale field studies comparing new nanoagrochemicals to their traditional equivalents would give a more realistic method of identifying the benefits of nanoagricultural applications in terms of efficiency, cost-effectiveness, and environmental impact. Besides, all nanoagrochemical-related research must include extensive characterization of both nano and non-nano products across a wide variety of application situations in order to build appropriate instruments for their regulation and associated advantages.¹⁹⁷

Nevertheless, it is necessary to educate the end-users regarding the importance of nanoproducts while strictly adhering to dosage limits (to ensure minimal side effects) and thereby fulfilling the purpose of using nanotechnology for sustainable agricultural development.

7. Conclusion

Recent progression in the field of agricultural nanotechnology has revolutionized the world with MONPs that could serve not only as a source of micronutrients but can also act as their carriers, improve the growth and productivity of crops, enhance tolerance against various biotic and abiotic stresses, and contribute towards agricultural waste, pest and disease management. In this review, we have mainly focussed on the ecological and cost-effective biosynthesis of MONPs using various biomolecules found in plants and microbes to develop biosynthesized MONPs, which can exponentially speed up their applications towards sustainable agriculture. Further, we have elaborated on different plant growth mechanisms, implying that MONPs increase mineral uptake, and photosynthesis rate, activate antioxidant defence system, increase stress tolerance, and upregulate growth-promoting genes. Besides we have briefly discussed about functionalization of MONPs and their successful application in agricultural waste management. The encouraging outcomes reported for the usage of biosynthesized MONPs as NFZs, NPDs, AMBs and NBSs suggest that they have promising potential to replace conventional agrochemicals when used in minimal dosages. Moreover, they are ultimately aimed at achieving sustainable agriculture *via* enhanced crop productivity/quality, crop protection from pests/diseases and detection of nutrients/pathogens/toxins in plants and soil. In conclusion, innovative developments of biosynthesized MONPs have great prospective for their implementation in eco-friendly nanobio-technology to drive the agriculture sector towards sustainability and enhanced crop production, protection and management.

Author contributions

Urvashi Gupta: conceptualization, writing – original draft, writing – review & editing. Dipak Maity: conceptualization, writing – original draft, writing – review & editing, visualization, supervision. Sumit Saha: writing – review & editing, visualization, supervision.

Conflicts of interest

The authors declare that they have no conflict of interest.

Acknowledgements

We would like to acknowledge the previously reported works by the researchers who have been cited in this review

paper. D. M. would like to thank University of Petroleum and Energy Studies for in-house financial support (SEED Funding: UPES/R&D-HS/24022022/08) and for all other supports. S. S. wishes to thank Prof. Suddhasatwa Basu, Director, CSIR-Institute of Minerals & Materials Technology, Bhubaneswar, India, for in-house financial support (Grant number: CSIR-IMMT-OLP-112) and requisite permissions. Finally, the authors would like to acknowledge Salonika Misra for her support.

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